

Physics of a Planetary Renderer

Models, approximations, and alternative approaches in PlanetSimulation

Implementation study · version 0.0.1

Abstract. PlanetSimulation couples terrain, prescribed orbital motion, reflected illumination, and an optically active atmosphere in an interactive renderer. This paper explains the physical ideas behind those models and the approximations that make them practical. Particular attention goes to atmospheric density, molecular and aerosol extinction, single scattering, twilight visibility, and refractive ray curvature. For each major subsystem, we compare the implemented method with a plausible alternative and state what physical behavior is retained or lost. Twelve generated vector diagrams explain the algorithms; versioned captures show actual rendering. Historical README timings are distinguished from new matched Quadro M1000M and RTX 3070 Ti measurements, with recorded inputs and workload limits. The model is intentionally limited: it does not solve climate, fluid flow, multiple atmospheric scattering, or mutual N-body gravity.

Keywords: real-time graphics · procedural terrain · Keplerian dynamics · atmospheric optics · adaptive rendering · reproducible engineering

1. Scope and physical questions

The paper describes the current source and its reproducible images, not a claim of physical fidelity for every planet. The useful questions are causal: what determines a surface point, what gravitational model moves the bodies, what radiance reaches the eye, and which terms can be approximated without destroying the intended visual behavior? Equations are simplified descriptions of the implementation; where a coefficient is tuned for the miniature scene, that is stated explicitly. Images come from `docs/screenshots/`, with capture inputs and hashes in `docs/captures/generation.json`. Diagrams in `figures/` are explanatory schematics generated by `make_figures.py`.

The atmosphere deserves special care. A convincing sunset depends on the length and density of the optical path, not merely on a sky color. The renderer therefore keeps absorption, out-scattering, in-scattering, and geometric visibility as separate quantities. Its reference pressure and temperature are configured inputs; no weather or climate equation evolves them. Tests and controlled captures check specific invariants and regressions, but an attractive image alone cannot establish that the numerical model is physically accurate.

2. Representing a planet in a camera

2.1. From configuration to a projected sphere

A position in a planet’s local mesh is transformed by a model matrix, a camera view matrix, and a projection matrix. In the project’s column-vector convention,

$$x_{\text{clip}} = PVMx_{\text{local}}, \quad x_{\text{ndc}} = \frac{x_{\text{clip}}}{w_{\text{clip}}}.$$

This mapping establishes which physical coordinates each calculation uses. Heights belong to the rotating planet frame; orbital positions and light directions belong to the world frame; depth and pixel reconstruction belong to the camera frame. Mixing them can make a mountain move under a stationary observer or make a shadow follow the camera. GLM supplies the matrix operations, while behavior tests check the resulting projections. Figure 1 records the data path.

From a configured body to a framebuffer pixel



Figure 1: The foundational transform and shading path. $P \cdot V \cdot M$ follows the current shader convention; the final display transform happens after linear-light composition. Generated schematic.

A hidden-window `--render-test` and PNG smoke test make camera framing, framebuffer readback, and image output observable in headless environments. Figure 2 is an actual capture of a compact regression scene. It is evidence of the current renderer at a fixed input state.



Figure 2: Headless rendering of the compact scenario. It checks the projected positions of the bodies at a reproducible input state.
Source: `docs/screenshots/solar-view.png`; provenance: `docs/captures/generation.json`.

2.2. A reproducible scene as an experimental unit

The JSON parser performs type and range checks before GPU resources are replaced. Scene replay stores a resolved scenario, camera, time, and quality choices next to a PNG. The image is an observation; its sidcar specifies a repeatable input state. On live reload, a new CPU scene is staged before the current one is exchanged, so an invalid or half-written file retains the last valid scene. This matters for physical comparison: changing the camera or exposure at the same time as atmospheric density would confound the result.

3. Terrain as a sampled physical surface

3.1. Procedural surface as a camera-independent function

The surface height is evaluated in a normalized planet-local radial direction u . A simplified form is

$$h(u) = h_{\text{landscape}(u)} + \sum_i A_i f_i(u; \text{seed}_i, \text{frequency}_i, \text{octaves}_i).$$

The implemented functions include value-noise fBm and a ridged variant. Broad continent and cliff fields determine large-scale shape; finer octaves add local detail. This is synthetic topography: it has spatial correlation and multiple length scales but no tectonics, erosion, or material transport. The seed and local coordinate frame make repeated samples of the same point deterministic. Since terrain geometry is rebuilt as the camera moves, a view-dependent height function would make the ground itself rise and fall. The current code samples `heightAt(radial)` during every LOD rebuild. Figure 3 distinguishes the fixed field from its variable triangulation.

Stable height field and camera-dependent tessellation



Figure 3: The height field is fixed in planet coordinates; eight distance levels change density and apply a bounded inward offset to coarser regions. Bars show the working segment anchors and interpolated levels; the sphere is schematic. Generated schematic.

The sphere begins with 20 icosahedral faces, subdivided twice into 320 stable base faces. For nearby views, each base face is assigned one of eight edge-segment levels using its angular distance to the eye. A previous-zone hysteresis margin prevents repeated switching at a threshold. Steep faces can receive extra segments if the triangle budget permits. Edges shared by unequal zones are sampled once at the finer requirement, then reused on both sides; concentric interior rings fill the faces without T-junctions. The budget loop first removes optional steep refinement and then lowers less-important zones. This preserves one continuous sampled surface under a per-planet triangle ceiling.

The working segment sequence, from coarse to fine, is 3, 5, 6, 8, 10, 12, 14, 16. Existing base, middle and maximum settings anchor levels 0, 3 and 7. Seven surface-distance boundaries span 300 to 1000 m. The three-zone counts in older reports remain an aggregate (level 0, levels 1–6, level 7); captures now also print all eight counts.

Coarse samples receive an inward displacement $s = s_{\text{max}} \left(\frac{n_{\text{max}} - n}{n_{\text{max}} - n_{\text{base}}} \right)^2$, clamped to zero for extra-steep detail. The configured `sink_depth_m` defaults to 1 m, additionally bounded to 0.1% of the radius remaining after maximum relief. Shared corners use the smallest incident offset; edges use the finer neighbor, with smooth interpolation toward corners and face centers. Shore refinement carries this scalar field through subdivision before displacement is applied. There is exactly one selected surface per direction, so coarse shells cannot poke through a simultaneously rendered fine shell. The sea uses the same eight-level selection with zero sinking. Geometry, shadows, reflections and foliage roots agree on the installed mesh.

This is a spatial sinking rule, not an animated LOD crossfade. Hysteresis and stable choices preserve vertices within unchanged levels; crossing a level can still move geometry. An offset below the sampled height also does not establish an error bound everywhere inside a triangle on arbitrary procedural relief. The accompanying `benchmarks/terrain-lod.md` records tests and controlled captures.

Two fixed before/after views (shoreline and grass) use all eight levels. Each already reaches the 100,000-triangle land budget and 60,000-triangle water budget, so this change redistributes detail rather than reducing those totals. Land uploads remain 12,000,000 bytes per rebuild; the sink scalar adds CPU metadata but no GPU vertex attribute or draw. The shoreline retains 19 finest base faces and the grass view retains 18. Capture

timings overlapped compilation and are not used as performance evidence. Replay inputs, image pairs, logs and hashes are retained under `benchmarks/terrain-lod/`.

An alternative is a cube-sphere quadtree with screen-space error bounds. It can allocate triangles more directly to projected error and support very large planets, but needs explicit seam and level-transition rules. A fixed high-resolution sphere avoids LOD transitions at the cost of drawing distant detail that cannot affect the image. Here the icosahedral base and fixed height query suit a small interactive scene where local walking must remain stable.

The sampled height feeds normals and terrain color factors as well as eye clearance. Water is a separate spherical shell, an approximation that omits tides and waves in the body's shape. The local camera follows the planet's translation and rotation through a planet-fixed frame, so the same latitude/longitude refers to the same terrain location after orbital motion or spin. Walking near a pole uses a great-circle step and parallel transport of the view basis; longitude itself becomes singular there. Figure 4 illustrates why transporting the basis in 3-D is better behaved than rebuilding it from longitude after every step.

Great-circle camera transport across a pole

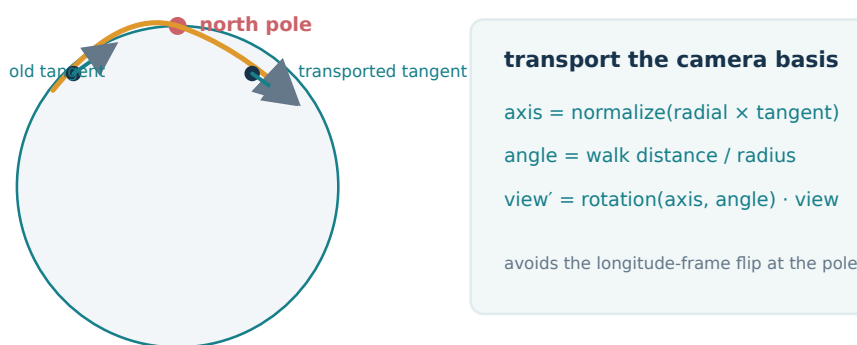


Figure 4: Great-circle position update and basis transport across the north pole. A longitude-derived frame can flip at the pole; a transported 3-D basis remains continuous. Generated schematic.



Figure 5: Surface-camera capture of procedurally generated terrain. Geometry derives from the stable height function while tessellation is selected for this camera. Source: current gallery.

3.2. GPU field evaluation and planned surface ownership

GPU field evaluation and complete consumer ownership now support opt-in captures; interactive acceptance remains pending. The starting source audit found CPU bulk height/gradient/material sampling and duplicated land vertices, CPU shoreline subdivision and grass planning, but already GPU density-column tables and atmospheric ray integration. The migration retains the 320 triangular icosahedral panels: CPU workers select topology, reconcile shared edges and choose inward sink scalars; GPU compute evaluates configured planet-local fields into reusable buffers. Hexagonal duals require pentagonal exceptions on a sphere, while H3 child boundaries only approximately fit parents. The checked design worksheet confirms twelve exceptional dual cells and manifold closure through three subdivision levels. A cube sphere is another possible future chart system, but changing topology would also change current face/replay identities.

CPU ownership	GPU ownership
Validated parameters and double body transforms	Bulk shape, gradient and material fields
Canonical topology, LOD and sink scalars	Resident terrain and water buffers
Sparse matching collision triangles	Grass metadata, roots and draw queues
Memory ledger and versioned installation	Atmospheric fields and integration

Figure 6: Generation contract. One complete generation supplies main views, shadows, reflections, foliage and contacts. Capture follows this contract; interactive compute acceptance remains pending and CPU fallback stays available.

The current land payload is 120 bytes per triangle. Assuming a closed, canonically indexed 100,000-triangle surface with 50,002 shared vertices, 32-byte double radial/sink descriptors plus indices would upload 2,800,064 bytes instead of 12,000,000: an analytical 76.666% reduction before parameters, metadata, alignment and staging. GPU outputs and spare generations still occupy memory; this estimate is neither a measured speedup nor a total VRAM saving. A bounded CPU collision mirror must intersect the same selected triangle planes and inward offsets as rendering, avoiding full-body GPU readback. The initial grass planner depended on CPU shaped vertices; T3b2 below removes that render mirror from new compute captures.

The plan keeps the existing GPU atmosphere table and optical integration, caches small uniform coefficient packs, and identifies the readback/sort of highlight tiles as a remaining reduction candidate. Future foliage budgeting reserves configured density in a near plateau before shrinking a Gaussian shoulder under measured memory/work limits; impossible near budgets require a stated fallback rather than an unachievable density promise. Backend/field/topology keys, canonical seams, double noise-coordinate arithmetic, GL resource limits, exact locked replay, sparse-contact coverage and atomic land/water/grass publication are prerequisite gates. GL 4.3 compute is opt-in until parity and total-cost measurements pass; GL 3.3 retains CPU terrain. Detailed ownership, panel comparisons, phased tasks and acceptance targets are in `architecture/terrain-gpu/plan.md`.

The first prerequisite (T1) separates a read-only CPU field oracle with a 704-byte parameter pack from indexed 32-byte radial/sink inputs. A bounded exact-coordinate height cache and validated field/topology/backend keys expose planning and bulk-evaluation work. Seventeen independent pre-refactor mesh snapshots retain exact vertex, normal, color, index and sink bytes, including both poles and production views. The implementation and validation are in `architecture/terrain-gpu/contracts/study.md`.

The compute proof (T2) generates double fields and legacy draw buffers on GL 4.3. A 100,000-triangle generation uploads 2,800,880 input bytes including parameters and controls instead of 12,000,000 shaped-mesh bytes (76.66% less). Packing, poles, extreme seeds, mixed LOD, shoreline seams, winding, replay and GL 3.3 fallback are checked. Land and water complete before publication. T3a now uses a radial topology index and lazy 1,024-position oracle cache for matching triangle-plane contacts, preserving float rounding and sinking without full mesh scans or GPU readbacks. CPU subdivision remains; grass compatibility vectors were retained through T3b1. Interactive compute awaits asynchronous consumer publication and hardware cost gates; no total-frame speedup is claimed. Studies are in `architecture/terrain-gpu/compute/study.md` and `contacts/study.md`.

T3b1 generates resident 64-byte grass triangle descriptors from the existing compute terrain buffers: original IDs, double bounds, conservative biome eligibility and Gaussian-weighted area. A 100,000-triangle oracle uploads 176 bytes and retains 6,400,160 additional bytes; native weighted-area error is at most 2.21e-14 on llvmpipe. Renderer readback and cached uploads are zero. Queried limits, chunking, buffer sizes, generation keys and GL range restoration are checked. At T3b1, CPU slot allocation/render vectors were retained; that prerequisite added work without claiming a frame speedup. The study and evidence are in `architecture/terrain-gpu/grass-metadata/`.

T3b2 now computes budgeted hash selection, rounded-slot density search and deterministic group/bucket prefixes on GPU. Fresh compute captures use `gpu-v1` and omit full CPU land/water render vectors and bulk field evaluation; older replays retain CPU planning. A 100,000-triangle case uploads 2,840 metadata/allocation bytes and reads one 224-byte capture summary, submitting 1,048,429 candidates within a conservative queried limit. Triangle/seed/rank root identities remain stable. No hardware frame improvement is claimed; T3c still owns asynchronous complete consumer installation. The implementation and validation are in `architecture/terrain-gpu/grass-allocation/`.

T3c1 introduces one persistent CPU executor with one replaceable queued snapshot and one completion slot. Requests own field/settings/topology inputs and carry epoch, body, field, mode and serial identities. The worker constructs canonical compute land/water topology and matching sparse contacts without bulk render evaluation. Reload invalidates old work without joining; shutdown joins before trace/context ownership ends. Normal CPU walking polls completed work and rejects incompatible or older generations. Compute capture explicitly waits for CPU preparation and retains its GPU waits; complete asynchronous GPU/grass publication remains T3c2/T3c3. Contact ownership temporarily duplicates canonical CPU topology during staging, so this checkpoint does not claim lower total memory or a hardware speedup. The study is in `architecture/terrain-gpu/async/worker/`.

T3c2 adds owned land/water and resident-grass preparation. GPU command order carries terrain-to-metadata-to-allocation dependencies without CPU fence waits during submission. Polling uses zero-timeout fences and retrieves one completed 224-byte allocation summary, validating prefixes, slot caps and candidate totals before preallocating texture views, blade queues, indirect commands and VAOs. A separate resource fence precedes patch commit. An initial 512 MiB logical ceiling reserves terrain and worst-case grass buffers; it does not estimate driver physical memory or available VRAM. Captures use explicit waits through these APIs at their original planning eye. Atomic scene publication, retirement and interactive opt-in remain T3c3/T3c4. The study is in `architecture/terrain-gpu/async/gpu/`.

T3c3a adds a complete transaction owner with off-live land/water/grass/contact resources, separate terrain and grass planning anchors, pre-dispatch overlap admission and no-throw publication transfers. One GPU preparation globally and one preparing/retiring spare per body bound overlap; default logical limits are 512 MiB per set and 1 GiB aggregate. Publication validates request and live consumer identities, creates a last-use fence before transfer, then keeps all old resources together until zero-timeout retirement polling signals completion. Disabled water and grass replace prior consumers at the same boundary. This owner is not yet connected to renderer frame boundaries or reload: T3c3b/T3c3c must refresh dependent caches and preserve the old scene through replacement readiness. Interactive compute remains gated. The study is in `architecture/terrain-gpu/async/publication/`.

T3c3b connects complete resident capture consumers to the pre-character frame boundary. CPU worker outputs stay off live resources while a scoped planning preview derives the exact chase eye from prospective sparse contacts. It restores motion, contact caches, cameras, input presses and replay state before GPU submission; trails and exhaust advance only in the real step after publication. Grass-only replacement exchanges the patch and receipt without replacing terrain or contacts. Changed land invalidates shadows and every publication invalidates whole-frame reuse. Managed scene passes validate keys/revisions before draws, skip independent patch preparation and expose consumed main, shadow, reflection, grass and water revisions. Capture waits remain explicit; transactional scene reload and asynchronous interactive acceptance remain T3c3c/T3c4. The study is in `architecture/terrain-gpu/async/renderer/`.

T3c3c1 adds a whole-scene replacement owner. A validated off-live CPU scene/config owns resident terrain, water, grass and contacts for every body. One exclusive replacement/retirement lease prevents competing terrain submission; aggregate admission charges old scene sets before replacement dispatch. Complete readiness and unchanged request/live identities precede a last-use fence and no-throw exchange of scene/config, mesh arrays, grass/trail state and receipts. Old resources remain owned and charged through zero-timeout retirement, including failed polls. Body reordering/count changes travel with their own fields and mounted cameras. Renderer reload integration and movement/body-switch acceptance remain T3c3c2/T3c3c3; interactive compute remains gated. The study is in `architecture/terrain-gpu/async/scene/`.

T3c3c2 connects the owner to `Renderer::reload()` and shared R/file-watch handling. A fresh config/replay snapshot and preallocated tracking arrays accompany all-body terrain/contact/grass preparation. Exclusive future-epoch CPU builds reuse the same executor without changing the live epoch on failure. Scoped prospective character previews restore old state; exact body-local replay anchors avoid coordinate round-trip rounding, and trail/exhaust data are validated before exchange. The complete boundary swaps tracking and replay/options,

rebinds cameras/contacts, resets character/effects and invalidates shadow/reflection/frame/orbit caches. CPU and legacy adapters stage the same consumers and fence old resources. One retired scene remains owned until last-use completion; captures wait explicitly, interactive CPU retirement polls without a timeout. Backend/planner, dimensions and renderer quality pipeline changes are rejected. Movement/body-switch/Moon acceptance remains T3c3c3 and interactive compute stays gated. The study is in `architecture/terrain-gpu/async/reload/`.

T3c3c3 extends character planning to prospective contacts for every body. Destination handoff binds the committed contacts immediately, and flight collision/chase queries use those triangle planes rather than waiting a frame or falling back only to an analytic height. Future-epoch reload preparation supersedes queued and ready old work while an executing snapshot retains ownership until completion and is discarded before replacement preparation. Failed replacement leaves the live epoch usable. One running, queued and ready slot remains the bound; compute reload waits belong to the capture adapter. Interactive opt-in remains T3c4 and hardware acceptance T3c5. The study is in `architecture/terrain-gpu/async/recovery/`.

T3c4a adds a resident frame core with zero-timeout GPU readiness and retirement polls. Startup and local-mask changes use the persistent CPU worker, and GPU backpressure leaves its single completed slot occupied. Complete publication refreshes tracking and contacts before character stepping and managed main/shadow/reflection/grass/water draws; grass-only failure retains the previous patch and contacts. Initial loading defers walking and scene draws while events continue. Failed preparation backs off unchanged input for sixty preparation frames, while incompatible work is discarded. Asynchronous whole-scene reload remains T3c4b and native input/public CLI acceptance T3c4c; CPU remains default. The study is in `architecture/terrain-gpu/async/interactive/`.

T3c4b prepares resident scene reload across frames. A latest immutable snapshot owns a monotonically increasing attempted epoch; nonblocking worker leases discard obsolete queued/ready results while executing snapshots finish with their ownership intact. Live publication is frozen during the scene lease, while old-scene motion and managed draws continue. One off-live body at a time retains meshes and sparse contacts rather than all full CPU outputs. A restored all-body preview supplies final grass anchors, then the shared complete exchange updates tracking, cameras/contacts, clocks and caches. Failed work retains the current scene; zero-timeout retirement keeps old resources until last use completes. Requests during startup or retirement wait without capture adapters. Native input/public CLI acceptance remains T3c4c, hardware cost T3c5 and CPU the default. Implementation and evidence are in `architecture/terrain-gpu/async/interactive/reload/`.

T3c4c enables explicit interactive compute with resident GPU grass while retaining the CPU default, fresh GL 3.3 fallback and locked replay rejection. Startup and reload share planner validation; enabled foliage must use compute placement. Native loading acceptance exposed a mesh profiler scope ending after its frame had been cleared. Loading presentation now follows scope destruction, and character stepping stays deferred until complete initial consumers are ready. A private executable observes the production CLI, visible GLFW window and interactive loop at presentation boundaries; actual X11 keys drive movement, while test-only GL hooks measure waits and control delayed/failed preparation and retirement. Product code gains no test observer or injected motion. T3c5 hardware frame-time and physical-memory acceptance remains pending. The study is in `architecture/terrain-gpu/async/interactive/native/`.

T3c5b1 adds optional trace-owned GPU timing for terrain field evaluation and corner expansion, grass metadata and ordered allocation, and main/reflection placement. Each receipt retains epoch/request/body/field/topology identities independently of staged resources; committed grass patches carry their originating request into later draws. A fixed 64-event pool checks both timestamp queries for availability, records dropped/missing samples and never forces GPU completion. Frame traces separately export the full start/end GPU span, readiness status and stage overflow. Placement intervals are nested in geometry stages and must not be added to render-stage totals. CPU default and hardware cost acceptance remain unchanged: complete native-loop wall time, publication latency/outcomes and physical-memory sampling remain T3c5b2, matched hardware experiments T3c5c. Design and validation are in `architecture/terrain-gpu/async/hardware/timing/study.md`.

T3c5b2a separately records complete native-loop wall time before event polling through the end of profiler collection/closure, including loading presentation and minimized-window event waits. A frame-linked CSV retains contained polling, presentation and idle-wait intervals with rendered/loading/minimized/exception outcomes. Stack-owned scopes close early loading continues and exception paths; disabled tracing reads no additional clock and this timer issues no GL calls. The original frame/GPU markers retain their boundaries. Contained intervals and GPU work overlap wall time and must not be added to it; loading/minimized samples need separate interpretation. Publication latency/outcomes and physical-memory sampling remain T3c5b2b/

T3c5b2c, followed by T3c5c hardware acceptance. Design and evidence are in [architecture/terrain-gpu/async/hardware/loop/study.md](#).

T3c5b2b1 records ordinary CPU/compute terrain and resident grass attempts from admission through the first complete consumer draw. A fixed 64-record scalar pool retains independent attempt IDs, phase offsets, generation/revision keys and wall/frame latency. Grass-only attempts retain the terrain serial but receive distinct IDs, also carried by GPU work. Failure, rejection, coalescing, cancellation, undrawn replacement and shutdown have explicit outcomes without successful latency. Worker callbacks only update metadata; CSV draining occurs outside the mutex and adds no GL wait. Reload transactions, captures, destination handoff and CPU grass-only endpoints remain T3c5b2b2, followed by physical memory and hardware acceptance. Scope and evidence are in [architecture/terrain-gpu/async/hardware/publication/study.md](#).

T3c5b2b2a extends the same bounded scalar pool to asynchronous resident reload roots and per-body worker/GPU children. A root starts before configuration validation; child GPU readiness describes off-live preparation, while final generation/grass receipts arm only after whole-scene exchange and live contact binding. A returned complete scene draw publishes matching children and then the root; empty scenes also require that draw. CSV parent, exchange-offset and child-count columns preserve the overlapping scopes. Invalid configuration, supersession, preparation/final-exchange failure and shutdown retain unsuccessful outcomes. Dropped children make the root trace incomplete rather than successfully measured. Completed-child drainage cannot lose root ownership; disabled tracing reads no trace clock and no GL wait is added. Synchronous CPU/capture reload, capture endpoints, actual handoff and CPU grass admission remain T3c5b2b2b; physical memory and matched hardware acceptance are still pending. Scope and evidence are in [architecture/terrain-gpu/async/hardware/publication/transactions/study.md](#).

T3c5b2b2b completes the publication lifecycle across synchronous CPU/legacy/resident reload, capture, actual destination handoff and CPU grass-only admission. Captures close terrain receipts after the returned complete scene draw; cached presentation does not invent consumption. Synchronous reload roots and children arm only after whole-scene exchange; failures or exchanges replaced before drawing remain unsuccessful. A handoff receipt observes actual destination contact binding through a matching character draw, excluding previews and flight travel time. Pure CPU grass begins only after rebuild admission on the existing mesh, recording CPU phases and its actual anchor; initial/replacement grass stays inside terrain latency. Admission-scoped capture mode and origin/binding-offset columns distinguish the overlapping intervals. Exception scopes close unconsumed attempts, while stable borrowed renderer members avoid topology retention or new workers. Unchanged legacy grass retains its originating GPU timing identity. Synchronous off-live work retains child identity between capture frames, with an empty GPU frame field and restored containing-frame binding. Replay setup that replaces an exchanged child before drawing yields an incomplete root rather than successful latency. Physical memory and matched hardware acceptance remain pending; CPU terrain stays default. Design and evidence are in [architecture/terrain-gpu/async/hardware/publication/lifecycle/study.md](#).

T3c5b2c adds a separate optional memory CSV. Exactly one nonzero OpenGL context UUID selects an NVML handle, whose UUID must read back identically before exposing device-wide physical bytes. One persistent worker owns initialization, 250 ms reads, process RSS, output and teardown; render observations use try-lock for latest state and a separate fixed 64-event queue with one producer/consumer. Lifecycle events survive latest-lock contention; skipped snapshots and queue overflow retain separate counters. NVX is queried only at context setup/shutdown. Rows retain query/metadata age, stale and unsupported/error statuses, reload attempt/epoch, logical overlap and partial stable CPU vectors. The replacement ledger already reserves the live scene, so overlap takes the larger reservation rather than their sum. Event rows reference cached physical readings and identify readings finishing after observation. Asynchronous startup may overlap later admission; periodic samples can miss transient peaks and include other processes. Teardown joins only outside the native loop; a stalled driver may delay shutdown. These diagnostics do not establish hardware cost acceptance or change CPU default. Plan and scope are in [architecture/terrain-gpu/async/hardware/memory/study.md](#).

T3c5c1 performs three alternating production native CPU/resident-compute stationary pairs on Quadro M1000M at 1280×720, with orbit/spin paused at time zero and wind active. All pairs match Earth's 100,000 triangles, 1,999,246 submitted foliage candidates, field/topology keys, camera/root and planning anchors. Effective density matches at 47.149 blades/m², below configured 120.72 under existing budget scaling; the protected near-density controller remains separate. Each run keeps 240 measured frames and full exclusions. Complete native p95 compute/CPU ratios are 0.9696, 1.0260 and 1.0492, close to the 5% limit in the least favorable pair. Primary observer overhead remains included and is not subtracted. Periodic device-wide used-memory maxima are 496.4 MiB on CPU and 521.9 MiB on compute; no replacement occurs. These receipts

establish production workload preflight, not migration acceptance. Actual 6/12 m/s routes, rendered near-root coverage, Moon/reload overlap and optimized bulk CPU field-time gates remain T3c5c2-c4. CPU stays default. Study and full provenance are in [architecture/terrain-gpu/async/hardware/cost/preflight/](#).

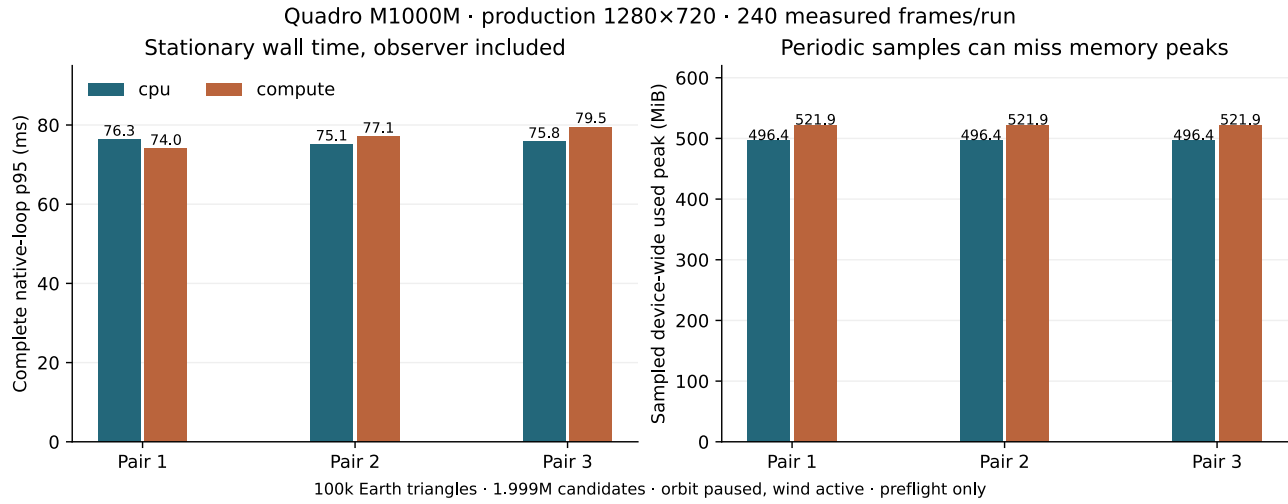


Figure 7: Production native stationary preflight, three alternating pairs. Complete loop p95 includes the private observer; physical memory shows periodic device-wide sampled maxima and can miss true peaks. All six runs retain matching scalar workloads. These are neither replacement-memory gates nor an accepted migration speedup.

T3c5c2a identifies a native grounded clock defect before matched route cost acceptance. The 50 ms per-frame cap produces correct 6/12 m/s animation rates but only 3.504/3.605 m/s walking and 7.028/7.167 m/s sprinting against monotonic wall time on CPU/compute Quadro production runs. Grounded movement now consumes elapsed time in camera/contact advances of at most 50 ms, bounded to one second after suspension. Extra advances use installed contacts while terrain/GPU preparation and publication remain once per frame; existing flight and jump activation behavior is retained. Natural 100 m production checks measure CPU/compute walking 5.978/5.974 m/s and sprint 11.906/11.926 m/s; adding 250 ms presentation delay still gives 5.971/5.975 and 11.938/11.971 m/s. All final speed checks pass $\pm 5\%$. An initial short fixed CPU sprint fails narrowly at 11.364 m/s because of a 91 ms observation-boundary difference; its full raw frames are retained alongside the longer checks. Fresh stationary compute/CPU native p95 ratios are 1.0368, 1.0227 and 1.0853; the third pair exceeds the 5% migration limit and requires further cost investigation. Longer three-pair routes, rendered near-root coverage, Moon/reload and final transfer/field/memory gates remain pending. CPU stays default. Full before/after receipts and retained qualification failures are in [architecture/terrain-gpu/async/hardware/cost/movement/](#).

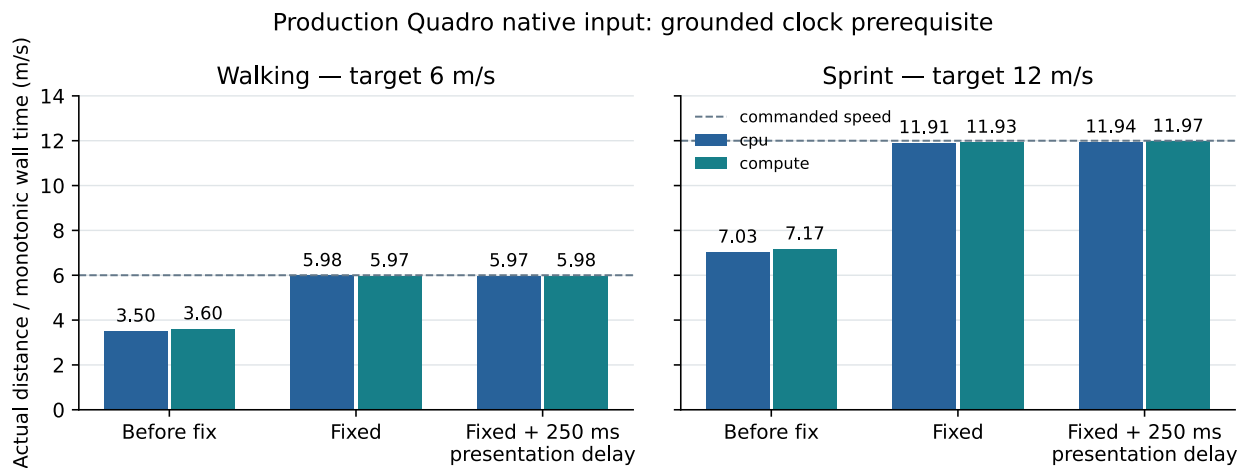


Figure 8: Actual grounded distance per monotonic wall second on production Quadro native input. Baseline uses short routes; fixed natural and delayed checks use 100 m. Each bar is one input run, with full raw spikes retained; these are speed prerequisites, not paired route cost acceptance or uncertainty estimates.

T3c5c2b1 collects three alternating native CPU/compute pairs each for 400 m walking and sprint on the unchanged production runtime. Common [5,400) m cohorts retain 5,276 measured and 3,797 excluded frames, including all 461 upper outliers. Walking is 5.825–5.990 m/s and sprint 11.948–11.996 m/s, within $\pm 5\%$ of command. Fixed 25 m landmarks match roots/cameras within 6.42/6.44 cm and retain actual nearest

generation/workload identities. Whole-route native p95 compute/CPU ratios are 0.7056–0.7312 walking and 0.5557–0.7045 sprint; small distance bins retain slower compute cases. Landmark effective-density ratios vary from 0.9710 to 1.0438. Sprint root-to-grass-planning-eye p95 is 25.15–25.28 m CPU versus 33.44–34.64 m compute, maximum 43.82 m compute. This includes the chase offset and is not a root-density measurement. The lower whole-route p95 remains cost preflight until independent rendered near-root coverage checks actual live generations in separate excluded inspection runs. Periodic device-wide peaks are 740.5 MiB CPU and 769.5–797.125 MiB compute; complete-run compute-owner logical overlap is 554.08 MiB, while the CPU ledger is unavailable. Sparse physical samples can miss transient peaks and include other processes. The stationary regression investigation, Moon/reload and final gates remain; CPU stays default. Method, inspection plan and full evidence are in `architecture/terrain-gpu/async/hardware/cost/routes/`.

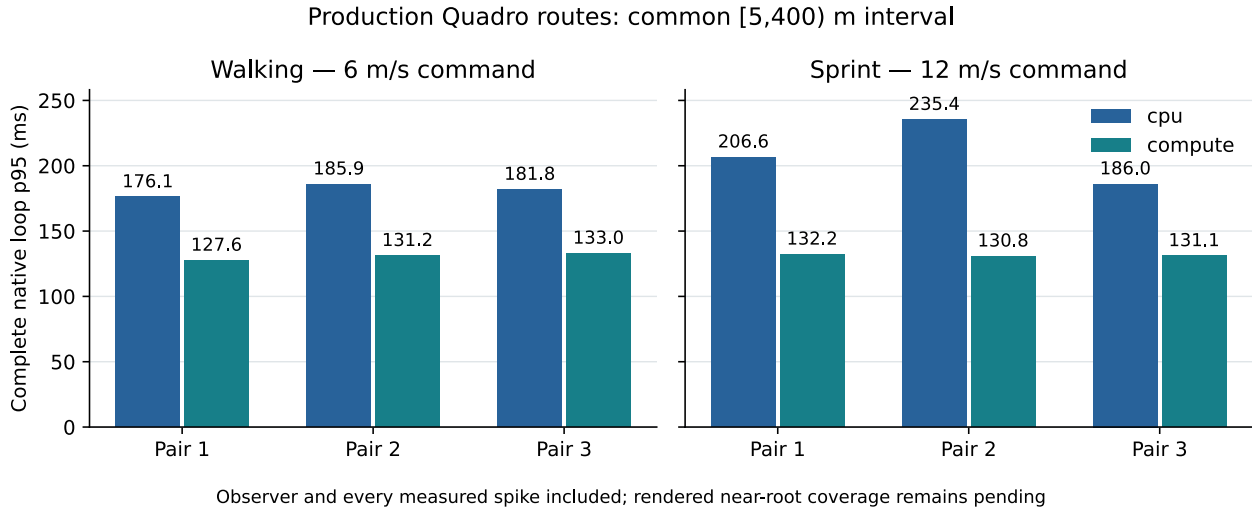


Figure 9: Complete native route p95 on production Quadro. Each pair covers the common interval from 5 m to less than 400 m at commanded walking/sprint speed. Observer overhead and every measured spike remain included. Rendered near-root coverage is pending; these bars do not establish accepted migration speedups or uncertainty bounds.

T3c5c2b2a adds a separate private native inspection executable, linking unchanged product libraries. It copies both 64-byte main-view Blade queue prefixes before reflection reuse and reads opaque depth before/after grass to export a pixel mask. Actual matrices, live terrain/grass keys and anchors, wind, astronaut pose and trail history remain attached to every snapshot. All synchronizing read calls/bytes are explicit and excluded from timing acceptance. Two final scoped software passes and eight Quadro fixture snapshots qualify the probe; two production sprint observations at 350.122/350.711 m retain 674,457/673,039 opaque grass pixels and distinct later reflection queue counts. These crossings have different wind/trail/pose state and much smaller anchor distances than the largest earlier lags. Similar raw counts cannot establish near-root density parity or dismiss the route discrepancy. Generated roots can be occluded; the opaque mask precedes character/water/atmospheric compositing and has no eligible-ground-area denominator. Those observations motivate the controlled T3c5c2b2b comparison below. CPU stays default. Source, complete evidence and next inspection controls are in `architecture/terrain-gpu/async/hardware/cost/coverage/`.

Live production sprint inspection: grass depth-change masks

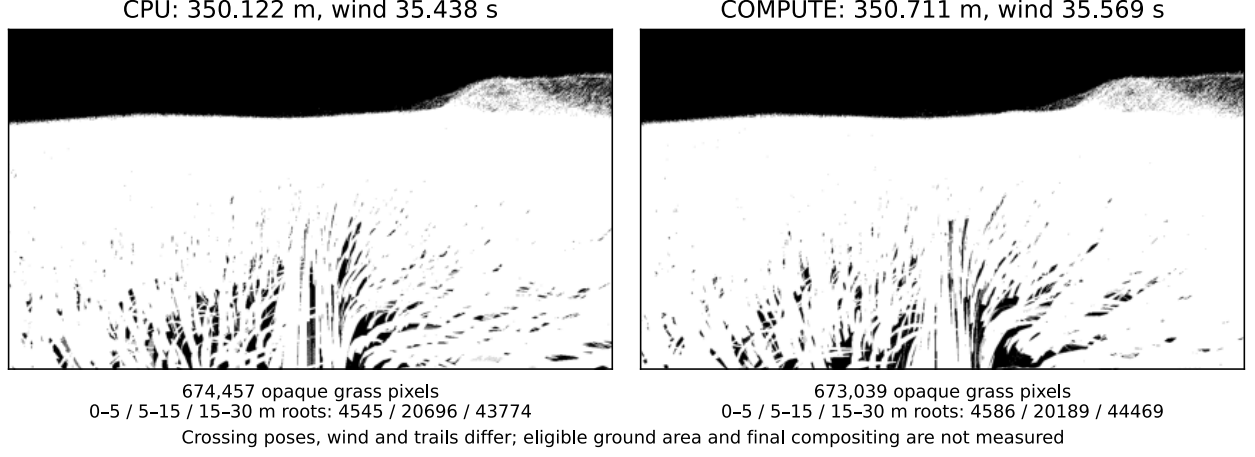


Figure 10: Illustrative actual native grass depth-change masks on production Quadro sprint. White pixels mark nearest opaque fragments changed by grass. Crossing pose, wind and trail states differ. Eligible visible ground area and final compositing are not measured, so these observations do not establish rendered density parity.

3.3. Matched live generations and estimator uncertainty

T3c5c2b2b fixes the inspection view, grounded pose, trails and wind while retaining each native installed terrain/grass generation. Private controlled scene renders skip CPU replanning; before/after publication, revision and workload receipts must stay equal. Full grass and suppressed-grass scenes retain character, water, atmosphere and tone mapping. Nearest opaque grass stencil plus final RGB difference supplies a composed contribution proxy; flare and overlays are outside scope. Live terrain rasterization records ground position, geometric plane, biome eligibility and surface-area weights. Projected roots use their own ray/triangle-plane intersection, avoiding pixel-center depth bias.

Twenty-four Quadro production routes supply three alternating pairs each for walking/sprint at first 25/350 m crossings. All 12,910 inspection frames are excluded from timing. Eight of twelve pairs pass the declared $\pm 5\%$ per-band area/root/fade/composed-coverage gate. Walking-25 pair 1 has near root/fade ratios 1.05362/1.05772 on identical ground but different topology/placement realization. Sprint-350 outer-band finite-quad area ratios 0.94390/0.93951/0.94340 fail. Actual shared roots are 25.546/25.048/351.020/355.518 m; native crossings can overshoot by 5.518 m, so these are common-view live-plan observations rather than exact interpolated route-cohort acceptance.

The area failures require methodological care. A finite derivative quad can exaggerate differential area near grazing triangle silhouettes. For positive camera depth z in metres, unit plane normal n , unnormalized camera ray d , viewport W, H and projection scales f_x, f_y , the analytic differential area is

$$J = \frac{4z^2}{WH|f_x f_y| |n \cdot d|}.$$

Post hoc evaluation on saved live planes gives outer-band compute/CPU ratios 1.01073/1.00559/1.01030. This identifies estimator sensitivity; it does not revise the declared gate or retroactively accept the samples. Known-plane/silhouette qualification, subpixel/band convergence and fresh frozen repeats remain T3c5c2b2c. Figure 11 distinguishes primary measurements from sensitivity results.

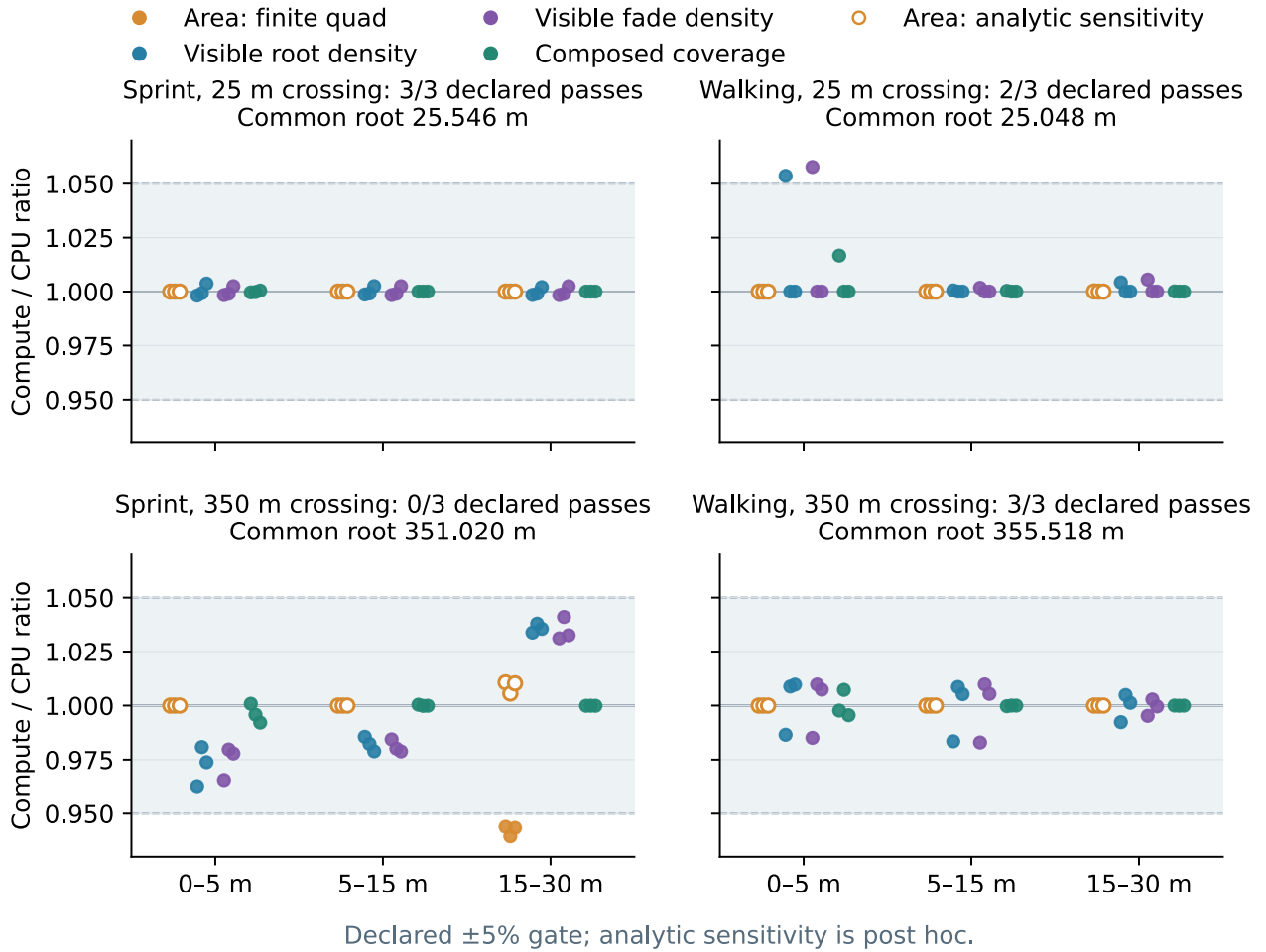


Figure 11: Declared common-view coverage ratios on three alternating pairs per crossing/case. Filled markers use the original finite-quad area proxy; open orange markers are post hoc analytic area sensitivity. Shading marks $\pm 5\%$. Eight of twelve primary pairs pass; no corrected-method or migration acceptance is claimed. Actual common root distances appear in each panel.

Three fresh stationary pairs using the unchanged ordinary timed probe retain 1,440 measured and 1,610 excluded frames, with native p95 ratios 1.05789/1.11058/1.04965; two exceed 5%. GPU opaque p95 is 4–6 ms higher, while grass placement means increase only 0.18–0.35 ms per view. Independent stage percentiles cannot be added as a whole-frame decomposition. Wind is active but its phase is absent from timed receipts; record/match it and isolate raster/overdraw cost before attribution or optimization. CPU remains default. Complete raw evidence, independent checks, declared controls and failure diagnostics are in `architecture/terrain-gpu/async/hardware/cost/coverage/matched/`.

T3c5c2b2c1 qualifies the analytic area on four known GPU geometries: front and 65-degree oblique planes, a foreground silhouette, and an 82-degree grazing silhouette with 1,000 m body scale, rotation and world translation. An exact float-rounded polygon/circle reference uses homogeneous frustum clipping and projected foreground halfplanes. The GPU rasterizes eligibility and planes into fixed tiles; excluded CPU analysis intersects original pixel rays and integrates 0–5/5–15/15–30 m bands. At a 640×360 base viewport, all software and Quadro scenes meet the declared 1% limits for 4× reference error and 2×/4× convergence. Worst errors are 0.34497% software and 0.33410% Quadro; worst convergence change is 0.42118%. A translated full-size viewport with unchanged projection bounds attachments to 2.25 MiB and meets a separate 0.01% tile gate. A failed coarse batch and one-pixel off-axis tile rounding failure remain archived. These known scenes qualify a measurement prerequisite, not live production eligibility, character/water occlusion, route coverage or timing acceptance. Live inspection qualification and frozen route repeats follow below. Details are in `architecture/terrain-gpu/async/hardware/cost/coverage/area/`.

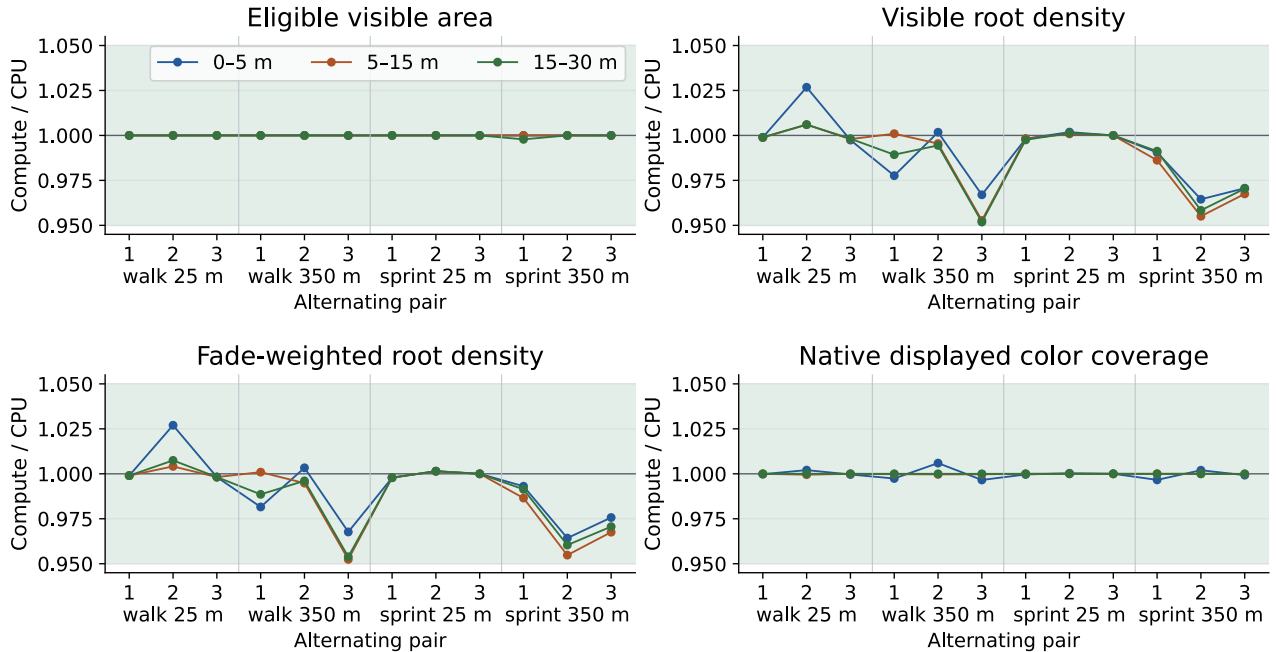
T3c5c2b2c2 qualifies excluded live analytic inspection on 20 CPU/compute snapshots: eight software fixtures, eight verified Quadro fixtures and four unchanged production walking-25/sprint-350 inspections. Installed land/other-body VAOs and full astronaut meshes supply independent opaque occlusion at 1×/2×/4×; fixed-projection tiles bound the new GPU target to 4.25 MiB. All 60 grids meet the 1% area-convergence gate: maximum change 0.78655% across fixtures and 0.27135% on production. Native 1× opaque tags match exactly;

180 separate biome diagnostics check recorded attributes and expected retention. Plans, anchors, revisions and publication/workload identities stay native, with complete character/trail telemetry restored. The native composed color proxy retains its display grid and a separately quantified edge-area uncertainty, reaching 13.45% of production near-band eligible area. This establishes neither 1% physical compositing accuracy nor native timing acceptance. Two single qualification pairs pass the 5% ratios; the required three alternating pairs across all four routes follow as T3c5c2b2c3. Coarse/schema/rebuild and close-handshake preflights remain archived; the private session now joins clean shutdown before checking its final trace. See [architecture/terrain-gpu/async/hardware/cost/coverage/live/](#).

T3c5c2b2c3 completes a fresh excluded route comparison after method qualification: three alternating CPU/compute pairs each walking/sprint at nominal 25/350 m crossings, totaling 24 verified Quadro inspections. All twelve pairs satisfy the unchanged $\pm 5\%$ per-band gates for eligible area, opaque-visible root density, fade density and native displayed color coverage, with at least 100 eligible pixels and visible roots per band. Maximum $2\times/4\times$ area change is 0.61349%; native opaque tags match exactly. The closest root-density ratio is 0.951834 in walking-350 pair 3's 15–30 m band. This accepts the declared fresh display-grid/root comparisons, not earlier failed samples, configured near-density preservation or timed route costs.

Adjacent complete native frames bracket every first crossing. Overshoot reaches 4.6648 m in walking-350, with a 5.9881 m bracket width and 4.4482 m CPU/compute native root separation. Common controls retain the first CPU view/pose/trail/wind for each case while plans, terrain/grass anchors and generation revisions stay native. Topology/field receipts and exact/rounded-millimetre raw root intersections record population variance; triangle/slot-based seeding makes topology dependence plausible, without establishing causation through a controlled ablation. The native color proxy retains edge-area uncertainty up to 13.45%; no supersampled physical blade-coverage bound follows. Stationary wind/raster investigation remains T3c5c2b3, and CPU remains default. Complete raw evidence and reproducible checks are in [architecture/terrain-gpu/async/hardware/cost/coverage/repeats/](#).

Fresh native Quadro coverage pairs; shaded region is the declared $\pm 5\%$ gate



Native plans and common controls retained. Color is a display-grid proxy with recorded edge uncertainty.

Figure 12: Fresh qualified Quadro comparisons: three alternating pairs for each nominal crossing/case, with native generations retained. All bands meet the shaded $\pm 5\%$ gate. Color measures a native display-grid proxy with separately recorded edge-area uncertainty, not physical supersampled blade coverage. Every route frame is excluded from timing acceptance.

3.4. Stationary wind and raster controls

T3c5c2b3 completes 54 native Quadro runs: three alternating CPU/compute pairs for each native/fixed/indexed wind and full/discard/suppressed grass draw control. The unchanged production workload retains 1280×720 , quality 1, 100,000 Earth triangles and a two-million configured/effective budget. Each run keeps

240 synchronized measured wind indices; all 12,960 measured frames, 18,320 exclusions and 426 upper outliers remain. Wind is recorded separately for main/reflection blade and placement shaders. Fixed wind uses 12 s; indexed wind uses $12 + 0.125i$ s for matched cohort index i . Native wind advances freely and is not phase-matched across runs. Hook/receipt observation remains inside complete native timing; blocking qualification queries are absent.

Fixed-wind full native p95 compute/CPU ratios are 1.094387, 1.150303 and 1.144083; all fail the original 1.05 limit. Their CPU/compute p95 values are 88.923/97.316, 86.197/99.153 and 86.932/99.457 ms. All fixed discard and suppressed-draw pairs stay below the limit. Indexed full pairs 2/3 also fail (1.068332/1.102773), while their ablations remain below it. Native full pair 3 and suppressed pair 1 fail, preserving background variability. Figure 13 shows every pair without filtering. Passing ablations do not accept full-workload costs.

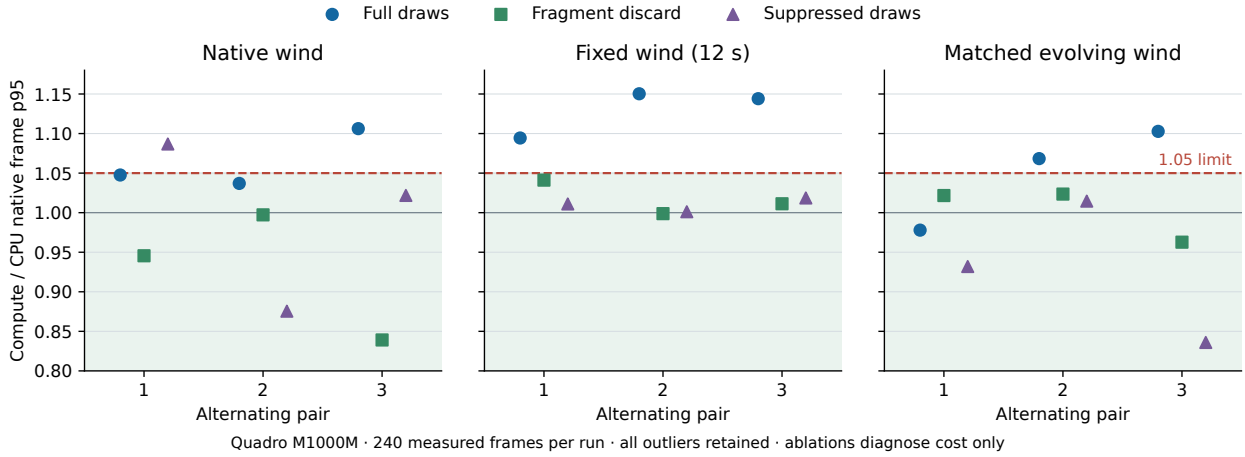


Figure 13: All 27 stationary Quadro pair ratios. Full draws execute the production grass path; discard retains placement/vertex work but disables rasterization; suppression omits grass draws after placement. The dashed 1.05 limit applies to full-draw costs. Ablations diagnose work and cannot qualify migration. All outliers remain.

Matching wind does not remove the repeatable regression. Discard and suppression localize it to work enabled by grass rasterization and downstream scene effects, without proving atomic order, root-population, overdraw or chunking causation. Fixed full GPU opaque p95 CPU/compute is 21.117/27.073, 21.842/27.118 and 21.783/27.082 ms; placement means increase only 0.210–0.311 ms per view. Reflection and atmosphere distributions also change. Removing grass changes their color/depth inputs, so independent stage percentiles and full-minus-discard p95 are not an additive frame decomposition or standalone fragment cost. T3c5c2b4 will distinguish the mechanism, apply a bounded correction and repeat affected full-workload pairs. CPU stays default; earlier failed samples, Moon/reload, protected near-density and final migration gates remain. Full evidence and reproducible checks are in `architecture/terrain-gpu/async/hardware/cost/raster/`.

A separate private blade-order probe qualifies the software controls needed for T3c5c2b4: stable native/ascending/descending camera-depth permutations preserve all 64-byte Blade records and indirect commands. Eighteen final CPU/resident snapshots retain exact shader matrices, opaque depth and displayed color, with populated detailed/quad queues in both main and reflection. Explicit blocking queue/pixel/query reads and restored roundtrip uploads are excluded from cost acceptance; every frame carries false timing/migration eligibility and captured frames join their own inspection receipt. CPU descriptors are distance-ordered while resident descriptors are triangle-ordered; physical-GPU cause remains unproven. T3c5c2b4 cannot yet collect native Quadro production evidence: direct NVIDIA device opens return EPERM, GLX/GLFW window creation and EGL initialization fail, while NVML inventory remains available. No runtime correction or new hardware performance result follows. Device access must be restored before diagnosis, correction and full-workload repeats. Raw evidence and resume commands are in `architecture/terrain-gpu/async/hardware/cost/raster/order/`.

4. Orbital motion and light transport

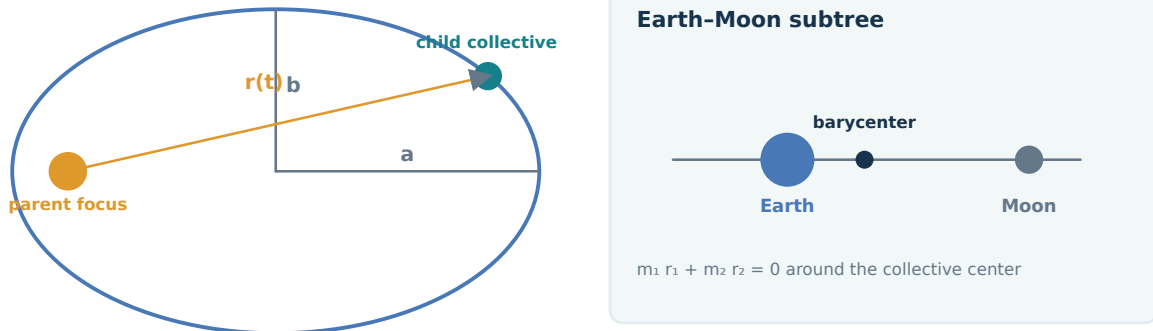
4.1. Prescribed Kepler ellipses with collective mass

A body names its parent and supplies semi-major and semi-minor axes a, b , a mean anomaly, inclination, ascending node, and periapsis orientation. The two-body approximation gives eccentricity and period as

$$e = \sqrt{1 - \frac{b^2}{a^2}}, \quad \mu = G(m_{\text{parent}} + m_{\text{collective}}), \quad T = 2\pi\sqrt{\frac{a^3}{\mu}}.$$

Here a and b are converted from the scenario's distance unit to metres before applying the SI gravitational constant; the code performs the equivalent calculation in world units. At time t , mean anomaly advances linearly, representing equal areas swept in equal times. A bounded bracketing solve obtains eccentric anomaly E from $M = E - e \sin E$, and the orbital-plane position is $(a(\cos E - e), b \sin E)$. Three rotations orient that ellipse in the world. The implementation uses a rationalized form near extreme periapsis to avoid cancellation. Evaluating at absolute time avoids frame-rate-dependent drift and makes pausing, replay, and speed changes deterministic.

Kepler ellipse and hierarchical barycentric recoil



Solve $E - e \sin E = M(t)$; then rotate the orbital-plane ellipse into world space.

Figure 14: An ellipse is specified relative to a focus; a child subtree and its parent recoil around a mass-weighted center. The right panel abstracts the Earth–Moon relation; sizes and offsets are not to scale. Generated schematic.

Each subtree acts as a collective point mass in its parent's prescribed ellipse; children induce mass-weighted recoil of the parent. Thus an Earth–Moon pair has an internal barycenter, and the Sun moves in response to its descendants while its configured position remains the time-zero anchor. This retains the main geometric and mass-weighted effects while omitting inter-branch perturbations and tidal evolution. A symplectic N-body integrator would capture mutual perturbations and long-term exchange of energy and angular momentum, but would need initial velocities, a suitable time step, and careful treatment of close encounters. Prescribed ellipses are a good fit for deterministic scene replay, not for predicting orbital evolution.

For the 0 overlay, the application samples the complete hierarchy at 321 times for each body: ten revolutions at 32 segments per revolution. A moon's inertial trail must inherit its parent's translation rather than drawing one stationary local ellipse. Figure 15 shows that distinction.

Ten future moon revolutions inherit parent translation

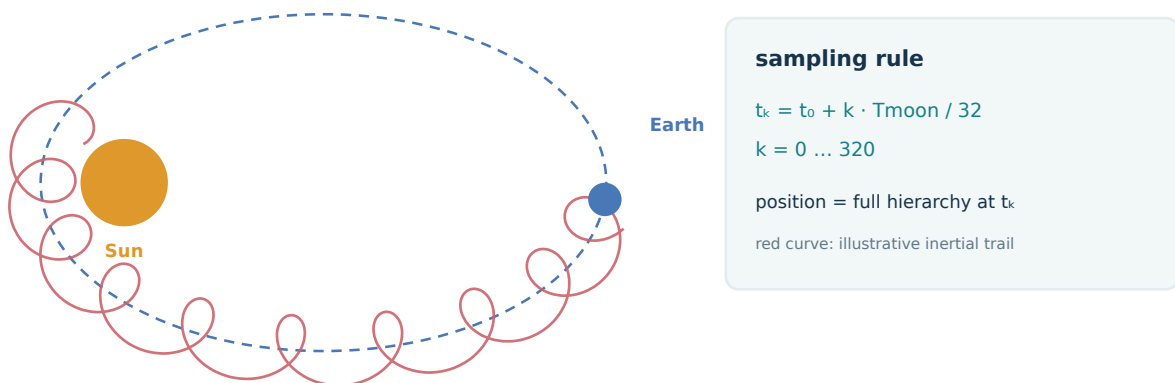


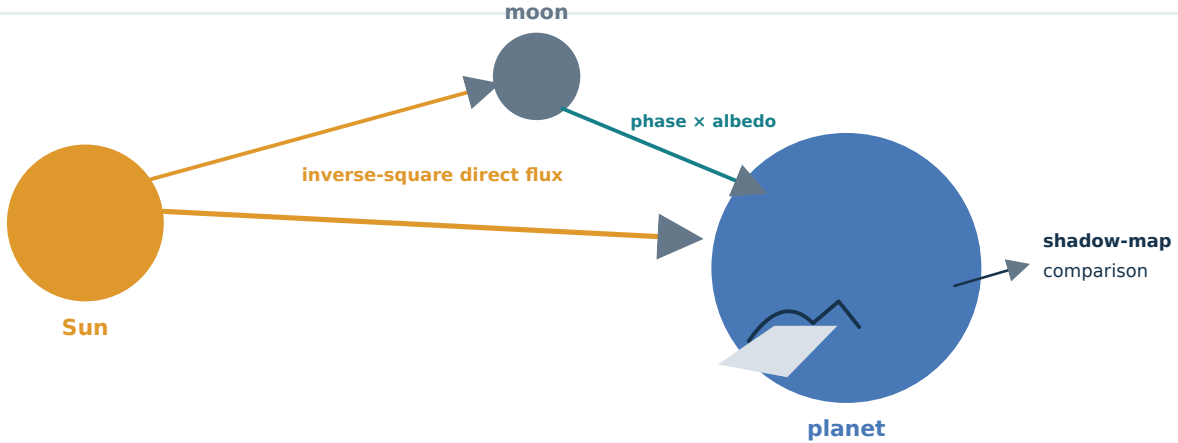
Figure 15: Illustrative moon trail in the inertial world frame. The repeated local cycles travel with the parent; the curve is explanatory and does not plot a measured scenario trajectory. Generated schematic.

4.2. Light transport before atmospheric composition

Per-frame CPU lighting computes direct flux with inverse-square distance and reflected flux with geometric albedo, a diffuse-sphere phase function, and inverse-square reflector-to-receiver falloff. Sun luminosity is derived from absolute bolometric magnitude using the IAU zero point. The phase law matters: a full moon presents more of its illuminated hemisphere to the receiver than a crescent. Shaders sum direct, reflected, and indirect contributions in linear RGB before an exponential display shoulder and sRGB encoding. Summing encoded display values would break the proportional meaning of flux and exposure.

A directional orthographic depth map is made per planet from the local Sun direction. The same map constrains opaque terrain, water, and direct atmospheric scattering. Its depth test approximates geometric occlusion by terrain; it does not solve light diffraction or soft penumbræ from the full solar disk. The map is cached until local Sun direction, terrain revision, extent, or resolution changes enough to alter its result. Ray tracing each point toward the Sun would give a more direct visibility test and potentially better soft shadows, but its cost is high for every surface fragment and atmospheric sample. Figure 16 separates the direct and reflected paths.

Direct illumination, reflected moonlight and terrain visibility



The same terrain depth map gates opaque shading, water reflection and direct atmospheric scattering.

Figure 16: The Sun contributes direct flux to each body, the Moon reflects a phase-dependent portion toward the planet, and terrain visibility controls direct light. The diagram is schematic; arrow widths do not encode flux.

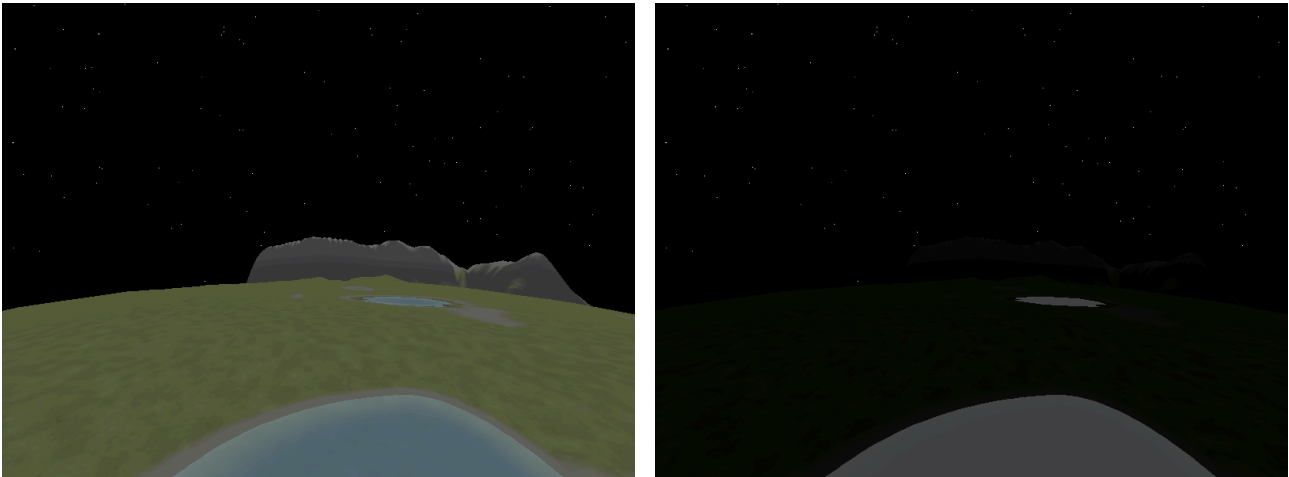


Figure 17: A controlled night-scene comparison from the current gallery: moonlit (left) and moonless (right). The saved replays preserve camera and lighting inputs, allowing the reflected-light term to be isolated.

5. Atmospheric optics: from air state to pixel radiance

5.1. Why density is modeled as a vertical profile

In an approximately isothermal atmosphere under roughly constant gravity, hydrostatic balance and the ideal gas law produce an exponential molecular density with scale height $H \approx k_B \frac{T}{mg}$ (R6). PlanetSimulation uses the same shape for molecules and a separate, usually shorter shape for aerosols:

$$\rho_{\text{m}(h)} = \exp\left(-\frac{h}{H_{\text{m}}}\right), \quad \rho_{\text{a}(h)} = \exp\left(-\frac{h}{H_{\text{a}}}\right).$$

This is a physically motivated shape, but the actual scale heights are set to fractions of the configured shell thickness, not calculated from local gravity and temperature. That choice keeps the visible atmosphere proportionate to a small rendered planet. Both densities taper smoothly to zero over the outer 15% of the shell so the finite numerical boundary does not appear as a hard optical edge. A hydrostatic solver with altitude-dependent temperature would produce more defensible densities, especially for a very deep atmosphere; it would also need gravity, thermal state, and boundary conditions that the current scene format does not supply.

At the reference surface, pressure and temperature scale molecular density approximately as $\frac{P}{T}$. Gas mixture changes the reference refractivity and Rayleigh strength. The code samples Rayleigh coefficients at three representative visible wavelengths and rescales their reference column from an 8 km Earth-like value into units of planet radii. This preserves recognizable color separation, including stronger short-wave scattering, but is not a spectral atmosphere prediction for an arbitrary small world. A spectral solver would resolve wavelength-dependent gas and aerosol properties more faithfully, then integrate to display RGB; it would require more storage and sampling per pixel.

5.2. Extinction, scattering, and the color of the sky

Molecular scattering, aerosol scattering, and aerosol absorption enter different terms. For a path of length s , optical depth and direct transmittance have the Beer-Lambert form

$$\tau(s) = \int_0^s \beta_{\text{ext}(l)} dl, \quad T(s) = \exp(-\tau(s)).$$

Here $\beta_{\text{ext}} = \beta_{\text{Rayleigh}}\rho_{\text{m}} + (\beta_{\text{aerosol-scatter}} + \beta_{\text{aerosol-absorb}})\rho_{\text{a}}$, evaluated per RGB channel. Extinction removes light from the original scene color; only the scattering coefficients contribute light into the view. In particular, treating all dust extinction as scattering would make a dense absorbing dust layer too bright. The renderer’s red-dust coefficients and droplet approximation are artistic optical parameters, while the molecular wavelength trend follows the Rayleigh idea.

The pixel combines attenuated scene radiance and light scattered into the view:

$$L_{\text{eye}} = T_{\text{view}} L_{\text{scene}} + L_{\text{in}}.$$

At each view-ray sample, incoming solar radiance is reduced by transmittance along the light path. The source is multiplied by local density and a phase function: Rayleigh scattering uses a symmetric $1 + \cos^2 \theta$ dependence, while aerosols use a forward-peaked Henyey-Greenstein form with fixed asymmetry $g = 0.65$. The source is then attenuated along the remaining view path before reaching the eye and contributes to L_{in} . This is why a long low-Sun path can redden transmitted light while a nearby sky region gains blue molecular scattering. A small indirect fill term approximates light arriving from other directions; it is not a multiple-scattering solve.

The aerosol state responds to configured dust and water. The water partial pressure is compared with a temperature-dependent saturation estimate; excess vapor can become suspended liquid according to a configurable fraction. A simple droplet cross-section gives an extinction estimate from liquid mass and droplet radius. Thus cooling at fixed water input can increase cloud or mist opacity in the model. It does not simulate nucleation, droplet-size distributions, precipitation, or convective transport. A weather model could supply local pressure, temperature, humidity, and liquid water to the optical layer; the current renderer consumes one configured reference state.

Terrain and the planet also control which sample receives direct Sun light. A spherical horizon test blocks the central solar ray below the horizon; a finite solar angular radius softens the twilight transition. Away from that narrow horizon region, the terrain shadow map can block sunlight from a sample near a mountain. The finite-disk transition is an approximation rather than an integration over the Sun’s actual disk, and shadow-map resolution limits the detail of mountain silhouettes. Without the terrain query, shadowed air would continue to glow as if its light path passed through a mountain.

The implementation uses bounded midpoint samples for the view integral. Within a sample it integrates constant source and extinction analytically through $\frac{1 - \exp(-\beta_{\text{ext}} ds)}{\beta_{\text{ext}}}$, avoiding an excessive bright contribution in thick dust. A precomputed multiple-scattering atmosphere, such as the approach in R2, would describe high-order skylight and horizon brightening more faithfully. It would require larger lookup tables and careful parameterization when

pressure, temperature, dust, water, or geometry change interactively. The present single-scattering calculation keeps those controls responsive and its cost bounded.

5.3. Why refractive rays bend

The same pressure, temperature, and gas composition produce an approximate surface refractive index. At fixed pressure, cooler air is denser, increasing $n - 1$ in this model. The radial profile decreases with altitude and tapers to $n = 1$ at the shell boundary. Fermat's principle says a ray follows a path of stationary optical length; in a smoothly varying index, its direction changes according to the perpendicular component of the index gradient:

$$\frac{d(d)}{ds} = \frac{\nabla n - (d)((d) \cdot \nabla n)}{n}.$$

Only the gradient perpendicular to the ray bends it; a gradient along the ray changes optical travel time but not direction. A midpoint step follows this curvature, and the radial symmetry gives a useful check: the spherical Snell quantity $nr \sin \theta$ should remain approximately constant. CPU tests check this invariant and ray reciprocity; GPU tests compare reprojected sky displacement with the CPU reference and check cold-versus-warm configured scenes. Figure 18 separates ray geometry from scattered radiance.

Atmospheric density, curved sightline and scattering

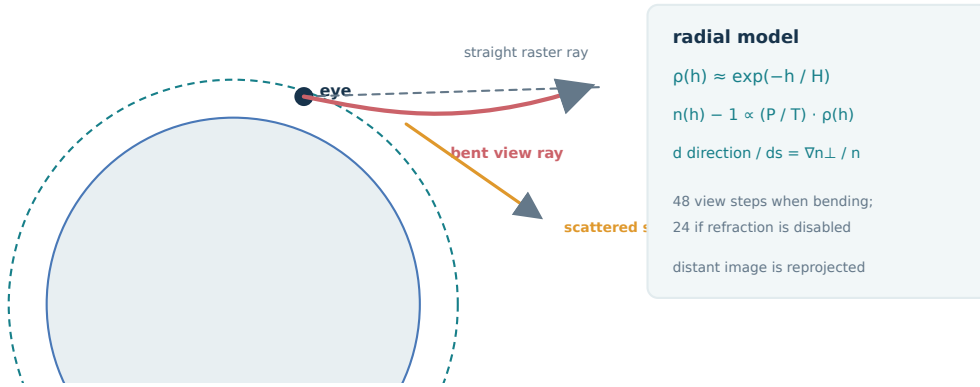


Figure 18: A refractive view ray bends through denser air while solar light contributes scattered radiance. Near terrain keeps its rasterized silhouette; only visible distant imagery is reprojected. Generated schematic.

The renderer bends a view ray and reprojects a visible distant image pixel when that ray escapes the shell. Near terrain retains its rasterized silhouette, and direct-light shadow rays remain straight. An off-screen or hidden source cannot be reconstructed from the existing color buffer; trapped rays do not leak background light. Full curved-ray tracing through scene geometry would represent refracted terrain and solar visibility more consistently, but would require a different visibility structure and substantially more ray queries. The pressure and temperature dependence has a physical basis (R3), while the chosen constants and miniature-world scale remain a rendering model rather than precision terrestrial refractometry.

A deliberately extreme test compresses the shell to 15 m around the 1 km planet, sets pressure to 150,000 Pa and temperature to 180 K, and places the surface camera 2 m above the ground with a 15-degree field of view. The two actual renderer captures below differ only in the JSON `refraction_enabled` switch. Refraction extends the visible solar limb into sky pixels near the horizon, while the already rasterized Sun and terrain keep their silhouettes. A small pointed edge remains where terrain hides the source image; this is a limit of image reprojection rather than a prediction of solar optics. The setting illustrates the rendering model's behavior, not Earth's atmosphere. The replay sidecars in `docs/captures/replay` preserve the exact resolved scenes.

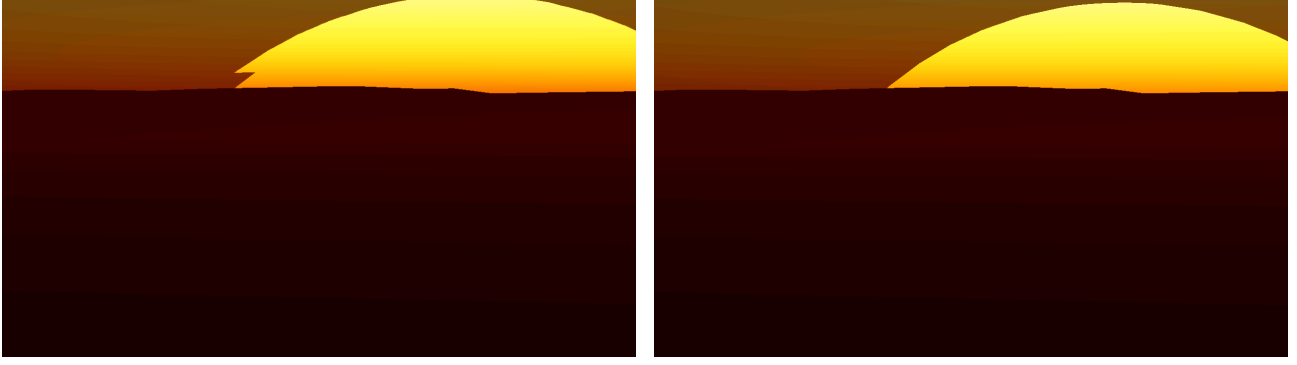


Figure 19: Matched full-resolution render captures of extreme refraction enabled (left) and disabled (right). Only the JSON bending switch changes; scattering and exposure remain active in both.

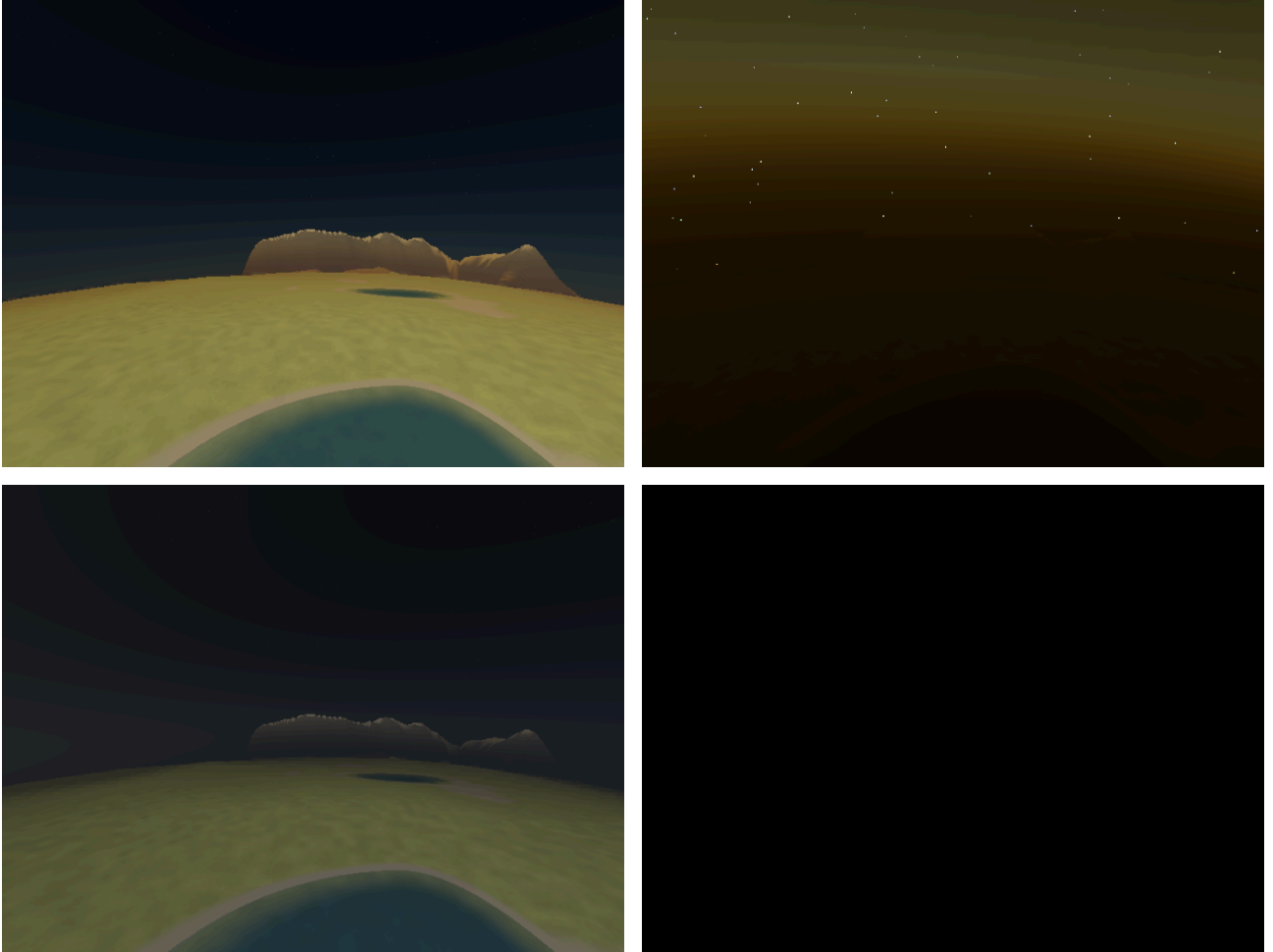


Figure 20: Current controlled atmospheric scenarios, left to right and top to bottom: standard air, sunset, condensed mist, and heavy red dust. All four are actual render captures from the regression gallery.

5.4. Where the atmospheric computation is spent

A scene-owned reference-air cache now supplies the same validated coefficient pack to sky lighting, body materials, main/reflection atmosphere passes and capture diagnostics. Its exact key includes composition, pressure, temperature, condensation/droplet settings, shell and physical radius, and enabled/refraction flags. Camera, Sun and body transforms remain dynamic. Each unchanged planet is evaluated once; reload builds an independent cache and invalid refreshes retain the previous pack. The small uniform calculation stays on CPU under the ownership plan. GPU density columns and integration remain unchanged; highlight tile readback and CPU sorting still await A1b. Design and evidence are in [architecture/atmosphere/optics/study.md](#).

For each visible pixel, the view ray needs repeated density and light-path evaluations. A reusable GPU density-column table stores the geometric integrals of molecular and aerosol density along straight light paths. Since the table stores columns rather than color, changing the gas mixture or dust coefficients can reuse the same geometry and apply new extinction coefficients. Camera motion also leaves the shell geometry unchanged. The main atmosphere field normally uses quarter resolution; a depth-aware resolve reconstructs smooth regions

and reintegrates full-resolution pixels near discontinuities. Terrain, stars, and the Sun stay at full resolution. This matches the physical structure of the calculation: smooth optical fields tolerate coarse sampling, while silhouettes and depth edges do not. Figure 21 summarizes the pipeline.

An alternative is full-resolution ray integration for every pixel. It avoids resolve artifacts, but repeats nearly identical work across neighboring sky pixels. A larger multidimensional lookup could reduce per-pixel work further, but would be harder to update for changing aerosol and humidity states. The current split keeps the expensive geometric column reusable while leaving composition as a live control.

Atmospheric composition: reusable columns and depth-aware fields

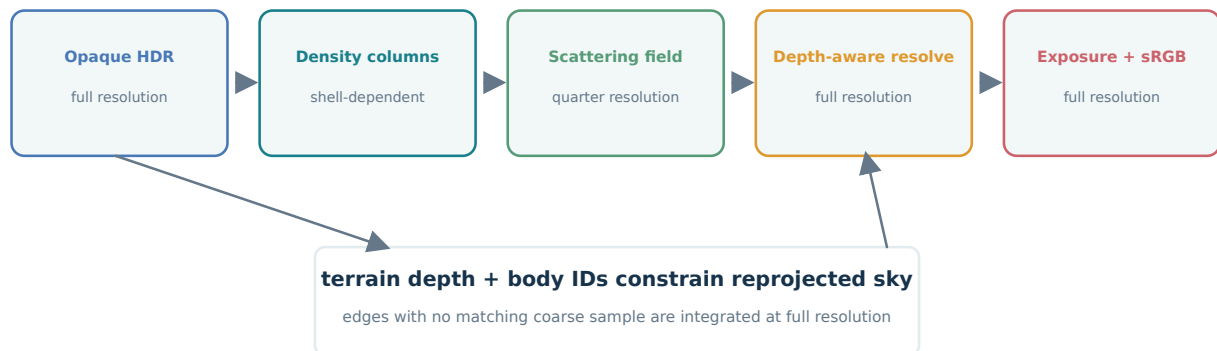


Figure 21: Atmospheric rendering graph. A reusable density-column lookup and a reduced scattering field lower repeated work; full-resolution depth controls edge reconstruction. Generated schematic.

The project's recorded Quadro M1000M benchmark at 1280×720 reported 208 ms mean frame time for the original full-field method, 138 ms with the column lookup, and 26 ms with reduced fields and shared ray calculations. These were 90-frame measurements of one surface scene with Debug CPU builds, recorded in the README; Figure 22 reproduces them rather than claiming a new experiment. The approximately eightfold change is specific to that workload and GPU. A separate RelWithDebInfo run reported 28.5 ms. A headless software-rendered capture is not comparable to this hardware benchmark.

Quadro M1000M benchmark reported in the project README

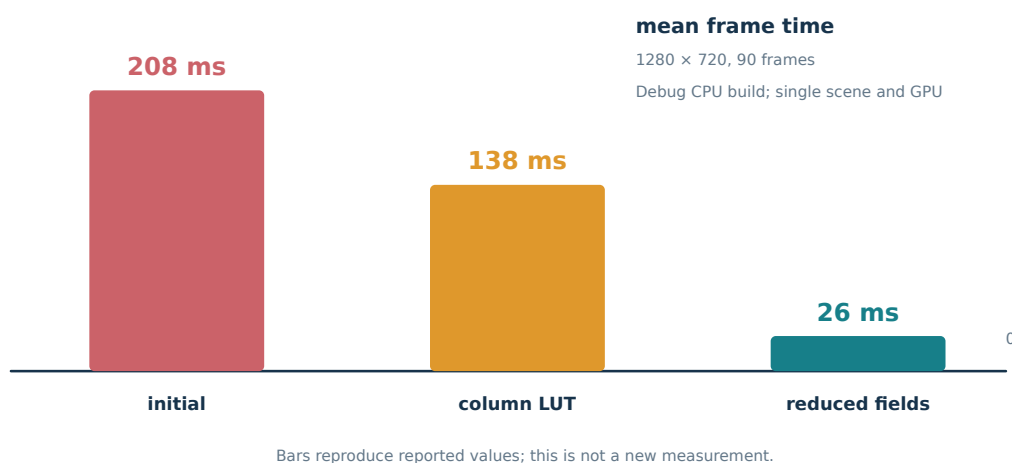


Figure 22: Reported mean frame times from README.md, Quadro M1000M, 1280×720, 90 frames. These bars visualize previously reported measurements, not a new benchmark.

A profiler now records CPU stage times and asynchronous GPU timestamp results without blocking for each query. CPU frame time, GPU pass time, and swap/presentation delay answer different questions; adding CPU and GPU times would double-count overlap. A completed HDR scene may be reused if camera, time, viewport, geometry, and pending-terrain state match. The HUD remains separately drawn so input can reveal or hide it immediately.

5.5. Matched physical GPUs: Quadro M1000M and RTX 3070 Ti

On October 5, 2026, both physical GPUs rendered the same production scene and unchanged RelWithDebInfo executable at 1280×720 through uncapped EGL pbuffers. Actual EGL UUIDs match NVML; both contexts are NVIDIA driver 580.178.04, GL 4.3 core, RGBA8/depth24/stencil8 with four samples. A private capture-only window/context adapter selects each card; renderer code, shaders, resources and profiler remain unchanged. Three serial pairs alternate Quadro/RTX, RTX/Quadro and Quadro/RTX. Each run renders 90 frames; ten warmup frames and the final readback frame are excluded, leaving 237 frames per GPU and case, all with valid asynchronous GPU samples. No completed scene is reused.

The initial surface camera uses production latitude -17.23072754436555° , longitude 79.39197457588234° , 2 m clearance and 80° FOV at simulation time 20 s. Time advances 1/60 s per frame. Walking also advances 0.1 m per frame, nominally 6 m/s in simulation time. CPU terrain and CPU grass planning remain default, with GPU foliage placement, quarter-resolution atmosphere, reflections and production shadows. Both cards plan 1,999,246 blades in 27,496 patches under the two-million budget and 150 m cutoff. Budgeted placement density is 47.1492 blades/m², below the configured 120.72 request; adaptive near-density acceptance remains pending.

Case	Quadro mean / P95 (ms)	RTX mean / P95 (ms)	Frame-time ratio
Fixed camera, orbital motion	73.279 / 73.606	5.745 / 6.078	12.75×
Walking, orbital motion	78.741 / 80.169	5.959 / 6.190	13.21×

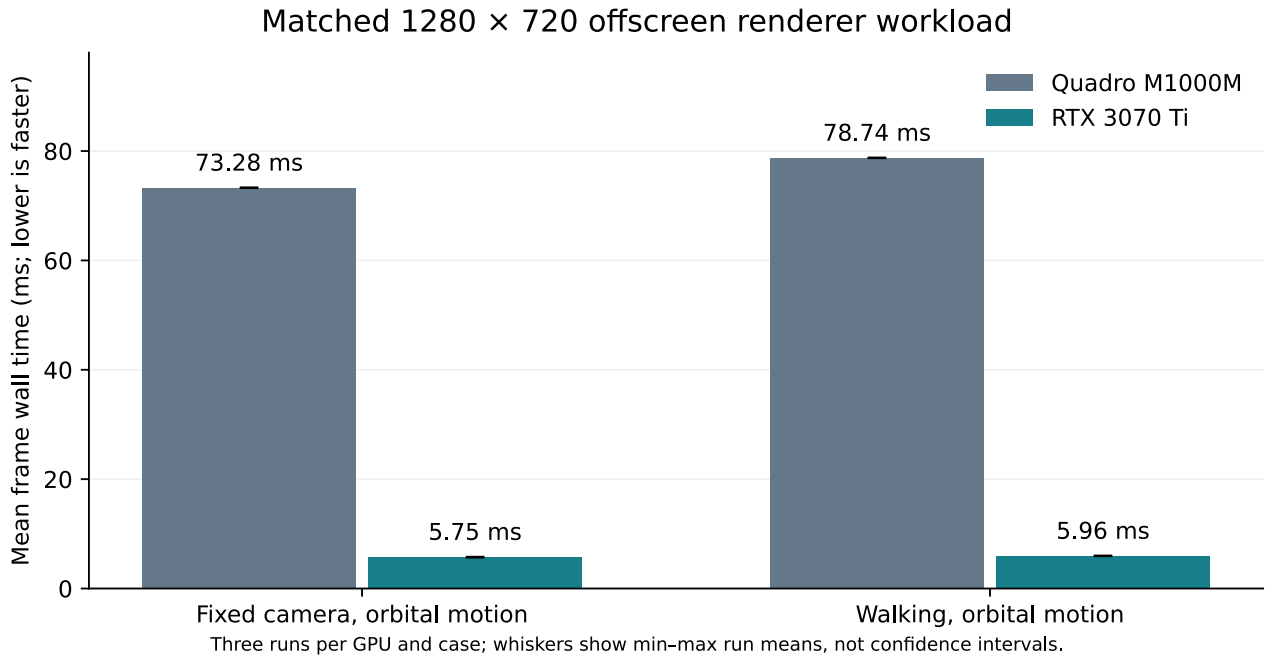


Figure 23: New matched physical-GPU measurements. Bars average three run means; whiskers show observed min-max run means, not confidence intervals. This uncapped offscreen comparison does not measure desktop presentation or native interactive FPS.

Fixed-camera scoped GPU stages total 73.038 ms on Quadro and 5.132 ms on RTX. Main atmosphere takes 33.630 versus 2.159 ms, opaque geometry 19.554 versus 1.247 ms, reflected geometry 11.741 versus 1.066 ms, and reflected atmosphere 7.215 versus 0.545 ms. Atmosphere and geometry remain the largest measured stages. These query totals are not complete GPU spans and must not be added to CPU wall times. Clocks and power limits were not locked; one-second telemetry is retained, with too few active RTX samples to establish sustained power or peak physical memory.

Camera and planned-count receipts match every pair. Main-view GPU culling differs by two blades (about 0.0011%), four triangles and eight vertices; paired PNG mean channel error is below 0.029 of one 8-bit level. Initial exact-count preflight was rejected; the final fresh runs require counts within 0.01% and mean image error below one level and record the actual differences. Neither case rebuilds terrain or grass during the retained interval: the short walk stays below the current cache threshold. Startup, longer movement, body switching and resident compute publication require separate cost measurements. This hardware comparison does not finish T3c5 CPU-versus-compute acceptance and is not comparable to the older, different-scene

README optimization timings. Full method, raw traces, all twelve captures, telemetry and frozen hashes are in `benchmarks/gpu-comparison/study.md`.

5.6. Pale sand with centimetre wind ripples

The sand reference requests pale grains and low wind ripples. Beaches now use linear RGB (0.86, 0.83, 0.75), independent of the planet tint, while their fragment-height boundary and grass placement remain unchanged. A body-fixed triplanar wave field has 18 cm spacing, a noise-warped phase and a bounded 7 mm amplitude. Grain at 180 cells per metre adds neutral colour variation, matte roughness and up to 0.7 mm additional height range. Screen derivatives convert this field into normal relief; projection-specific footprint filtering removes unresolved waves and grains. Geometry, depth, collision and shadow silhouettes retain the original mesh. This models the appearance of centimetre dunes rather than transported sand or geometric dune silhouettes.

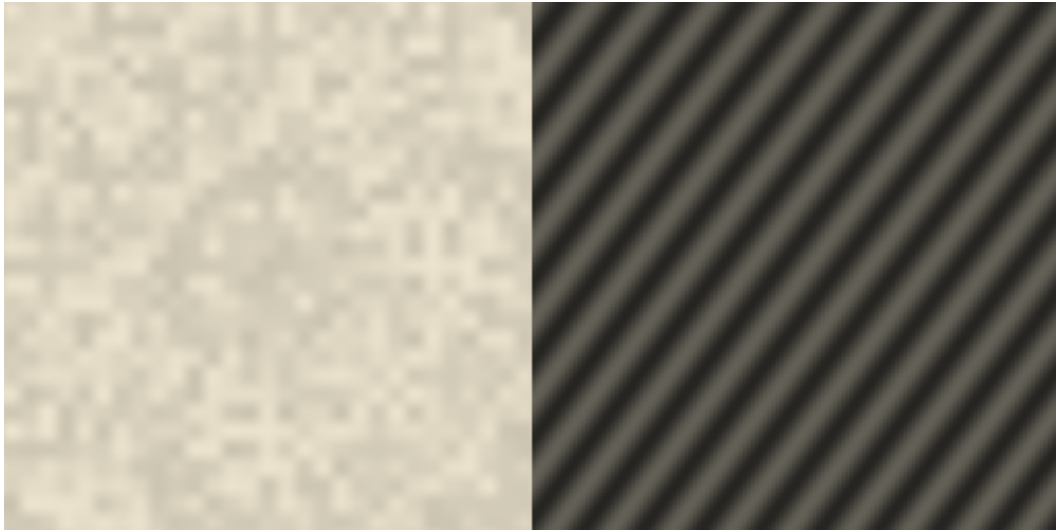


Figure 24: Production-shader diagnostic: fine grain across a 16 cm patch under indirect light (left), and wind ripples across a 2 m patch under grazing sunlight (right). Two triangles per patch, with unchanged depth; scales and lighting differ deliberately.

GPU checks cover pale colour under a green planet tint, narrow beach boundaries, grain and ripple visibility, distant filtering, rotation, both poles and unchanged depth. The retained shoreline comparison and validation are in `materials/sand.md`. The additional two value-noise samples and three projected waves increase fragment work without increasing geometry or transfers; no hardware timing improvement is claimed.

5.7. Perlin wind on a rotating planet

Grass wind now samples three advected 3D Perlin gradient fields, replacing the earlier overlapping sine waves. The field input is each root in body-local metres, so orbital translation and rotation do not change the local deformation. Gust cells span 12.5 m, direction cells 50 m, and flutter cells approximately 1.4 m. The implementation blends gradient dot products at eight cell corners with the quintic fade $6t^5 - 15t^4 + 10t^3$, following the construction in Ken Perlin's improved-noise reference. Integer hashing selects corner gradients instead of a permutation table. This is a visual wind model, not a fluid simulation.

The lattice repeats every 256 cells. Drift rates are binary fractions of a cell per second with a common 8192-second period, allowing the double wind clock to wrap before conversion to a float uniform without a discontinuity. Both the main and water-reflection pass use the same field and phase. Wind strength zero leaves the seeded static lean. Live wind follows wall time while T pauses orbits and spin; fixed captures preserve their explicit phase for deterministic replay. The vertex shader adds no textures, triangles or per-instance uploads, but performs more arithmetic than the old sine field. GPU transform-feedback checks exercise lattice and period boundaries, both poles, large wrapped times, fixed blade roots and independence from body transforms; pixel tests retain wind motion, shadowing and reflection clipping checks.

5.8. Historical grass geometry budgets and eight distance levels

These measurements describe the earlier single-triangle/sinking experiment. The current procedural quad implementation below supersedes its geometry, sinking and per-root transfer.

The glvertexid heightmap example reconstructs regular-grid coordinates from vertex IDs and connects strips with degenerate vertices. Its published example contains neither eight LODs nor sinking; those requested

techniques are adapted here to the existing instanced grass. Terrain retains its irregular spherical tessellation and shared shoreline edges, which cannot be represented directly by that regular height-only grid.

Blade strips now terminate in one tip, removing the former duplicate tip vertex and degenerate triangle. Eight nominal distance levels use 6, 5, 4, 3, 2, 1, 1 and 1 segments. At the working 60 m draw range, boundaries are 7.5, 9.375, 11.25, 13.125, 15, 30 and 45 m. Geometry straightens between 3.75 and 7.5 m, before any segment reduction can occur; removing collinear intermediate vertices preserves its shape. The CPU subtracts the 9 m movement allowance when choosing detail, keeping enough geometry while the cached patch follows the walker. Each retained blade submits no more vertices than the previous near/far scheme.

Eight deterministic retention groups thin distant candidates further, retiring one eighth on average in each equally spaced interval from 15 to 60 m. Smooth sinking and collapse update in the shader on every frame. CPU rejection uses the same group hash and adds the movement allowance, so a discarded root cannot become visible before the next rebuild. This deliberately trades distant density and shape detail for less work; the near-camera Gaussian candidates and terrain remain unchanged. Both main and reflection passes use the same decisions. One sorted instance buffer replaces two separate buffers. Compatible one-triangle levels merge into a single draw, leaving at most six instanced draws per pass rather than two previously, but only one instance upload per rebuilt patch.

Initial frozen grass view	Previous	Eight LODs
Uploaded blade instances	162,828	146,510
Vertices submitted per pass	1,477,562	1,081,766
Triangles submitted per pass	1,151,906	788,746
Instance bytes per rebuild	6,513,120	5,860,400
Instanced draws per pass	2	6

These counts come from the same generated roots: vertex submissions fall by 26.8%, submitted triangles by 31.5%, and instance payload by 10.0%. Triangle counts include submitted degenerates and fully sunk reserved geometry; bytes exclude driver overhead. CPU tests check guard-band visibility and the vertex bound. GPU transform feedback checks endpoint agreement, sinking continuity and distant shape invariance; a primitive query verifies submitted triangle accounting. Full timings, raw traces, executable and shader-tree hashes, frozen inputs and reproduction commands are in `docs/journal/benchmarks/grass-lod.md` and its data directory.

A paired 640×360 walking capture on Mesa llvmpipe with two worker threads measures 20 frames, excluding three warmup frames and the final readback frame. Mean frame time falls from 1576.97 to 1463.22 ms (7.2%); the median falls from 1387.17 to 1295.95 ms (6.6%). The foliage-disabled control changes from 708.99 to 720.05 ms mean, and grass preparation per rebuild remains similar. This single software-renderer pair measures a quality/performance tradeoff, not a hardware FPS guarantee. Both versions rebuild grass six times and install three terrain meshes in the measured window; bare final captures match exactly.

5.9. Procedural grass and terrain-buffer reuse

Grass now swaps between six-segment strips (14 vertices, 12 triangles) and low-detail tapered quads (4 vertices, 2 triangles). A narrow top edge preserves both low triangles. Current camera-to-triangle bounds select geometry without rebuilding roots, while the fragment shader evaluates a shared nonlinear palette so low detail retains the same base-to-tip color. Dithered pixel coverage replaces sinking; grass roots and heights stay fixed while coverage retires. The landscape retains its independent eight-level sinking.

The JSON `quad_distance_m` setting selects the morph endpoint in metres, with morphing starting at half that distance; zero preserves the automatic minimum of 15 m and one quarter of the grass draw distance. CPU draw selection and both shader paths share the threshold independently of root density. Interactive wind follows wall time independently of orbital pause and speed, with whole-frame reuse disabled while wind is animated. Fixed captures retain their explicit wind phase for reproducibility. The native `GrassWindInputIntegration` test waits for the completed surface transition, presses T and the speed keys, verifies changing grass pixels at fixed orbital time, reloads a zero wind speed and requires identical RGB frames. Removing foliage proves that the frozen view contained grass. Resolved scenes, images and traces are retained under `benchmarks/wind/`; see `benchmarks/live-wind.md`.

The quad transition now preserves interior shading by interpolating lateral offset and normalizing by local blade width per fragment. A neutral-light regression reproduced a 13-level RGB difference before the fix.

Candidate acceptance uses a stable slot rank with a smooth density-coverage ramp, so crossing a power-of-two batch boundary no longer replaces existing visible roots. Compute/fallback position parity and both transition regressions pass.

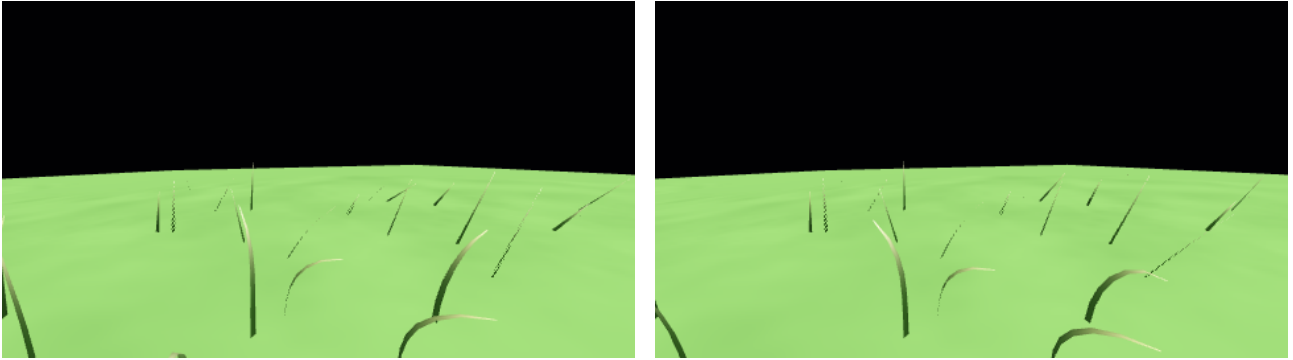


Figure 25: Native surface frames one second apart with T-paused planetary time. Grass wind continues in the same body-local field; a separate zero-speed comparison is byte-identical. Controlled scene disables water, atmosphere and overlays.

Grass reuses the terrain’s GPU vertex/index buffers through OpenGL 3.3 buffer textures. It uploads only one four-byte triangle ID per selected patch plus small rule uniforms. GLSL generates barycentric roots, blade variation, Gaussian acceptance, biome rejection and Perlin wind. No individual grass positions, blade meshes, random arrays or noise fields are uploaded; cached frames upload zero grass buffer bytes. A bounded CPU plan still reserves candidates and draw ranges. OpenGL 4.3 computes placement and wind once per candidate, compacts visible blades into two GPU queues and issues indirect draws without runtime CPU readback; OpenGL 3.3 retains procedural vertex generation. The separate horizon tuft layer and its controls were removed at the user’s request. Ground grass now uses the shared foliage-tip palette at midpoint variation (linear RGB 0.634375, 0.74375, 0.284375). This trades GPU evaluation for lower traffic and does not by itself guarantee higher FPS.

The JSON `foliage.wind_noise` block controls gust, direction and flutter frequencies in cycles per metre, an integer field seed and the wrapped clock’s speed multiplier. No sampled wind data crosses the CPU/GPU boundary. Full implementation details and collected validation are in `docs/journal/benchmarks/procedural-grass.md`. A frozen 100,000-candidate view uploads 8,666 four-byte triangle IDs (34,664 bytes), with 12,800,032 bytes of GPU working storage. Its main pass draws 8,678 detailed blades and 2,306 quads. A serial eight-frame, 640×360 walking pair on llvmpipe (five retained frames) measures 3125.60 ms mean with vertex generation and 2792.21 ms with compute, a 10.7% reduction; final PNGs match exactly. The main atmosphere pass still costs about 923 ms, and a synchronous capture terrain rebuild costs about 3.3 s. Reflection geometry is slightly slower with compute (208 versus 176 ms). Priorities are atmospheric work, sharing grass evaluation across passes, and terrain tiling or GPU generation: the existing land mesh transfers 12 MB versus 34.7 KB for its grass plan. These are software measurements, not hardware FPS claims.

The live-wind checkpoint subsequently passes all 50 CTest entries in one clean 333.81 s run on the same host. Its native frames change 11,178 RGB channels while 89 completed frames share orbital time 20.2094061 s; zero-speed frames are identical and reload remains paused at 20 s. Evidence is in `benchmarks/wind/validation/`.

5.10. Offline capture: bounded detail without a frame-time target

The `--offline-render` command captures a resolved surface camera at requested native resolution. An independent in-memory profile expands the single grass radius by 1–20 times (20 by default), uses full-resolution atmosphere, raises terrain edge segments to 32 under a 100,000-triangle ceiling, and uses a 256-segment Sun sphere. Other bodies, including a distant Moon, receive a ten-segment base; the camera planet retains its original coarse base to reserve triangles for nearby terrain. Offline sinking is zero. The source scene remains separately stored in the sidecar, so replay applies the profile once. Candidate growth follows area but automatic growth caps at two million (about 256 MB of compute queues); larger explicit scene budgets are retained. Density and biome rejection still apply. This sacrifices a frame-time target while retaining finite memory and tessellation bounds.

A controlled airless capture expands radius from 30 to 600 m and raises submitted Sun triangles from 2,048 to 131,072, Moon triangles from 960 to 86,400, and the camera planet from 9,964 to 99,732. Its 409,600 reserved grass candidates stay within the derived budget; they are not all visible blades. Production transform feedback on compute and vertex paths verifies visible roots beyond the normal cutoff. Actual scene coverage remains conditional on budget, Gaussian density, view and terrain suitability.

A final OpenGL 3.3 pass adds a solar halo, streak and four colored aperture ghosts along the Sun–image-center axis. Synchronous stencil readback measures depth-tested Sun visibility, and visible pixel brightness attenuates the contribution through atmospheric extinction. Fully terrain-hidden Suns contribute zero; partial visibility scales an approximate projected-disk area. The pass adds display-space color after tone mapping and preserves depth/stencil. This is an artistic lens approximation, not spectral lens ray tracing or calibrated diffraction. Its own `lens_flare` trace stage includes the accepted readback cost. Normal/offline on/off captures, zero contribution behind terrain, explicit override checks, invalid replay rejection before window startup and byte-identical self-contained replay are documented in `benchmarks/offline-rendering.md`.

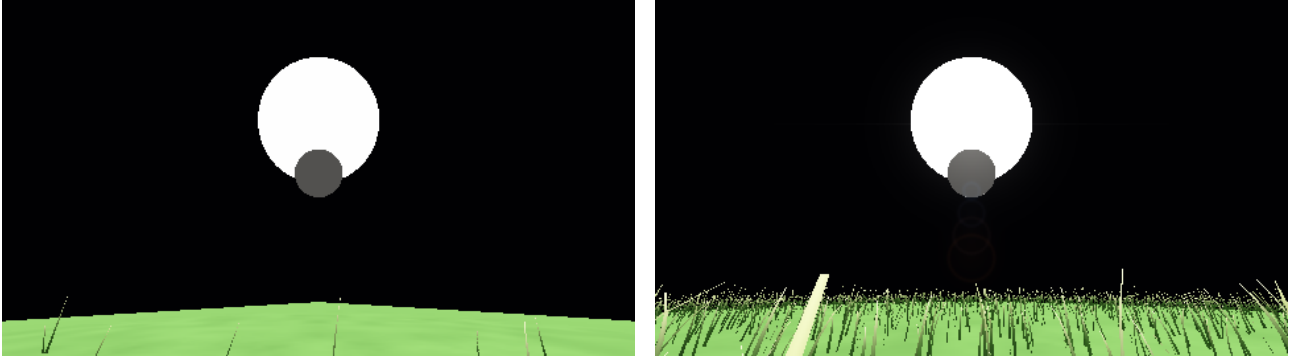


Figure 26: Controlled airless view: normal radius/detail (left), offline 20× radius with finer bodies and lens flare (right). The Moon partly covers the Sun. Images show a quality profile, not a matched-density performance comparison.

The later README showcase retains the production initial surface location, direction, 2 m clearance and timestamp 0, including its atmosphere, water and lighting. At native 1920×1080, an explicit 12-million-candidate foliage budget, 0.05 Gaussian sigma fraction and 65,536 per-triangle slots produce 213,454 main-view submitted blades, versus 102,918 with the production foliage settings at the same camera. The 20× profile gives a 3 km cutoff and 150 m sigma. Its two compute queues occupy about 1.54 GB; hard budgeting and biome rules still reduce density. The backlit dusk foreground follows the initial production lighting.

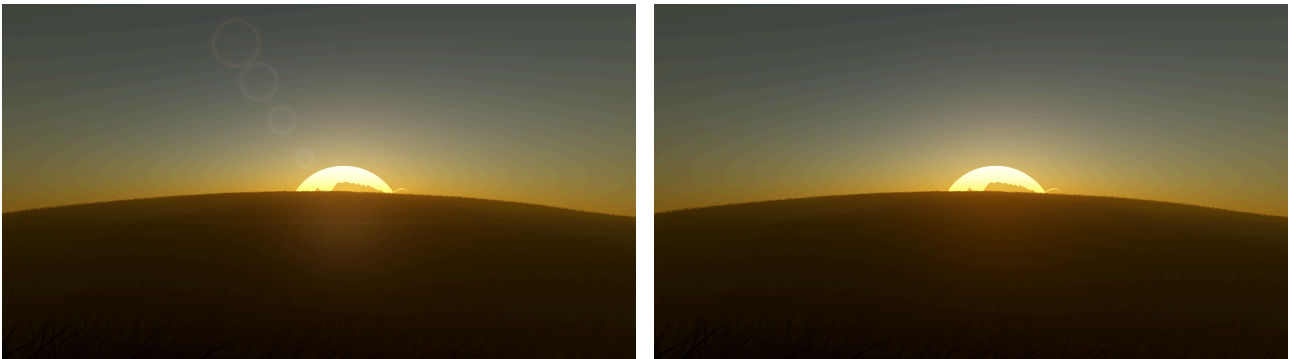


Figure 27: Production-start dense offline view: stronger, spread aperture ghosts and solar streak (left), identical scene with flare disabled (right). The dense PNG replays byte-identically. These are direct native-resolution renderer captures.

A central 192×1080 probe submits 43,690 blades but retains the same candidate queue bytes and 100,000/86,400 Earth/Moon triangles. Current terrain and foliage plans depend on the eye rather than each column’s frustum; narrower targets therefore cannot themselves load more geometry. Column assembly needs view-aware streaming with shared terrain boundaries, off-axis projections, stable root identities, one full-frame highlight/exposure solution, reflection coverage and a global flare pass after assembly. The current task records that decision instead of implementing a tiled image without allocation savings. Details, measured counts and replay/flare checks are in `benchmarks/offline-showcase.md`. Capture statistics now snapshot main-view queues before water reflections overwrite them, without adding interactive readbacks.

5.11. Camera movement and grass preparation: measured costs

The following measurements predate the Perlin wind change; their recorded executable hashes and captures identify the earlier sine-wind workload.

On 2026-09-30, a fixed-step benchmark separated the cost of moving the surface camera from orbital motion. Three cases share the same camera and scene: stationary with paused time, walking with grass, and walking without grass. Each capture has 36 frames at 640×360; five warmup frames and the final readback frame are excluded. Walking advances 2 m per frame. The working scene uses a 60 m grass draw distance and a 200,000-

blade budget. The host runs Mesa 22.3.6 llvmpipe with two worker threads under Xvfb; CPU code uses GCC 12.2.0 with `-O2`. These are software-renderer measurements, not a reproduction of the reported 30-to-10 FPS change on a physical GPU.

Case	Frame mean / median (ms)	GPU passes mean (ms)	Grass per rebuild (ms)
Before: stationary	36.64 / 36.59	2.09	0
Before: walking	1089.03 / 872.44	924.25	79.92
Before: walking, bare	620.87 / 478.28	481.32	0
After: stationary	37.26 / 36.96	2.20	0
After: walking	1033.55 / 869.35	878.71	50.79
After: walking, bare	639.14 / 482.17	498.60	0

Both walking runs rebuild grass on ten measured frames and install five terrain meshes. All thirty stationary frames reuse the completed scene. Captures construct terrain synchronously for reproducible geometry; live walking already constructs land and water in the background. The optimized capture’s CPU mesh stage averages 136.34 ms per frame, with a median near zero. This average must not be interpreted as a render-thread stall in the interactive path. CPU and GPU times overlap, and swap time includes software-driver waits.

Two local changes reduce grass preparation without changing its roots or appearance. A conservative triangle bound rejects candidates whose random sample cannot pass the Gaussian density test, before computing their full geometry. Sorting caches each squared distance once, then sorts compact keys and gathers the instance buffers. Mean placement time on rebuilding frames falls from 55.59 to 35.65 ms, sorting from 23.51 to 14.26 ms, while instance submission stays near 0.85 ms. Complete preparation falls from 79.92 to 50.79 ms, about 36%. Three isolated placement runs per version, alternating order and discarding two warmup iterations per run, reduce median process CPU time from 52.505 to 34.6775 ms (34%). All iterations retain exactly 162,828 blades and identical attribute checksums. All three paired final PNGs are byte-identical.

The median walking frame remains nearly unchanged, at 872 versus 869 ms. Rendering grass, reflected geometry and atmosphere dominates this software-renderer workload. Grass is already instanced and distance-cached; it is drawn in both opaque and reflection passes. Partial patch updates could reduce preparation spikes further, but would not eliminate those draw costs. They also require stable terrain tile identities: candidate seeds currently depend on triangle indices, which can change with tessellation. Reusing roots across mesh revisions without resampling could leave grass above or below the new ground. A future tiled cache or background placement job should retain overlap, rebuild changed tiles and discard obsolete jobs. The later culling and LOD work should measure each render pass on the target GPU before choosing a broader redesign.

Reproduction commands, complete per-pass tables, frozen inputs, raw CSVs, executable hashes and caveats are retained in `docs/journal/benchmarks/camera-movement.md` and its `movement/` data directory. Live traces now expose foliage placement, sorting, upload and rebuild counts separately. No end-to-end hardware FPS improvement is claimed from this single paired software-rendering run.

5.12. Nested CPU tracing and function-level profiling

The optional `--cpu-trace` records nested scopes with elapsed wall time and, where available, calling-thread CPU time. Render and terrain-worker lanes are separate. The standalone HTML report provides expandable caller trees, a chronological SVG timeline, self-time rankings, and reversed callee-to-caller paths. Self time subtracts immediate children; it still includes uninstrumented calls, driver work, waits, and tracing overhead. Inclusive times must not be added across ancestors and descendants. Recording begins in `Renderer::run`, after constructor startup.

A frozen 60 m grass scene was measured on 2026-09-30 with GCC 12.2, Xvfb and two llvmpipe workers, at 640 × 360 pixels. Fourteen frames advance the walking camera by 2 m per frame with simulation time paused; the first three and final frame are excluded. The ten retained walking frames average 4216.18 ms, versus 4179.30 ms with CPU tracing disabled; their final PNGs are byte-identical. This single pair includes host variation and does not establish an overhead bound. Stationary paused-image reuse averages 131.36 ms; walking without grass averages 2021.34 ms. These are software-renderer diagnostics, not hardware FPS measurements.

Scope over ten walking frames	Wall (ms)	Thread CPU (ms)
Grass draws, main + reflection	16059.27	16050.88
Atmosphere, main + reflection	18452.14	36.81
Terrain construction	4313.86	4312.70
Shoreline refinement (inside terrain)	3683.14	3682.22
Grass placement	421.21	421.17
Grass LOD batching	150.42	150.37
Grass upload	7.99	7.99

Grass draws consume about 38.1% of scoped frame wall time and 72.1% of the render thread’s CPU time, including software OpenGL work. Atmosphere takes 43.8% of frame wall time but little calling-thread CPU time, indicating blocking or scheduling while other driver workers may execute. Terrain construction contributes 10.2%; shoreline refinement is 85.4% of terrain time. Placement, batching and upload together contribute 1.4%, about 145 ms per rebuild. Capture terrain construction is synchronous; the interactive path uses a worker, so capture costs are not interactive render-thread stalls.

The independent Callgrind run records 7,473,993,658 instructions for one terrain build followed by twelve placement/batching iterations, including benchmark checksums. Its exported self and caller/callee reports identify placement, terrain noise, shoreline ordered-map comparisons, rounding and sorting as construction costs. Instruction shares are not time shares; optimized source attribution can split one function across several file entries. This focused CPU workload excludes rendering. Kernel sampling was restricted, and GNU gprofng warned that its interval timer changed; its inconsistent sampled times were rejected. A suitable target-GPU host is still needed for reliable function-level sampled CPU times and hardware optimization conclusions.

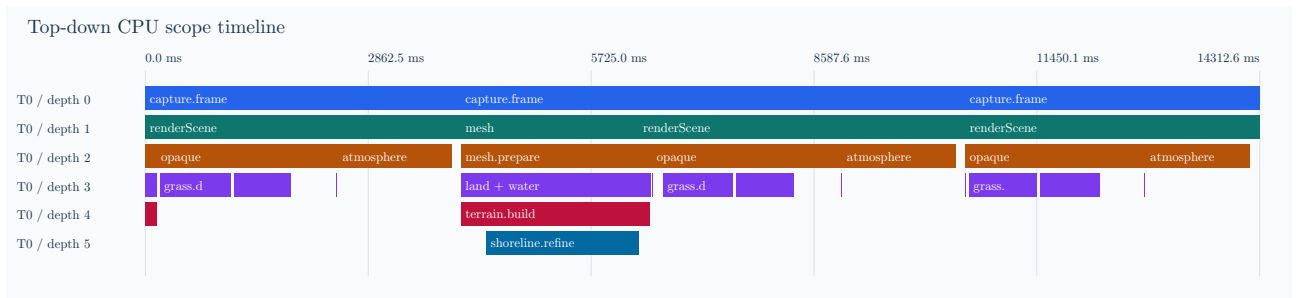


Figure 28: Top-down nested CPU scope timeline for three frozen walking frames on llvmpipe. Width represents wall time, including waits.

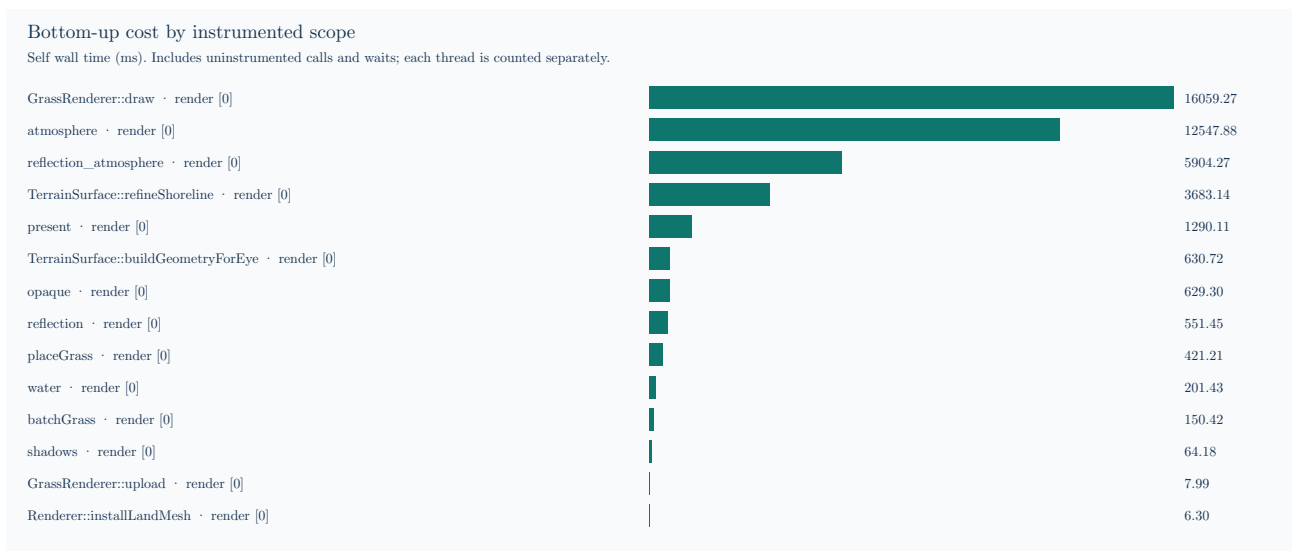


Figure 29: Bottom-up self wall time over ten walking frames. Immediate child scopes are subtracted; uninstrumented work and waits remain.

The retained study is described in `docs/journal/benchmarks/cpu-profiling.md`. It motivates investigating rendered grass/reflection work, atmosphere execution and shoreline construction before assuming that partial placement updates solve the frame-time problem.

6. Visibility, exposure, and interactive constraints

6.1. Diagnostics, dependency boundaries and adaptive quality

The I/Tab panel reports timings and optional Linux NVIDIA NVML utilization. Utilization is sampled off the render thread and is device-wide, while GPU pass milliseconds belong to this application’s measured passes. Software rendering and ambiguous GPU matching show N/A; a fake driver library tests the ABI path without claiming validation on NVIDIA hardware. Figure 30 is a real captured panel.

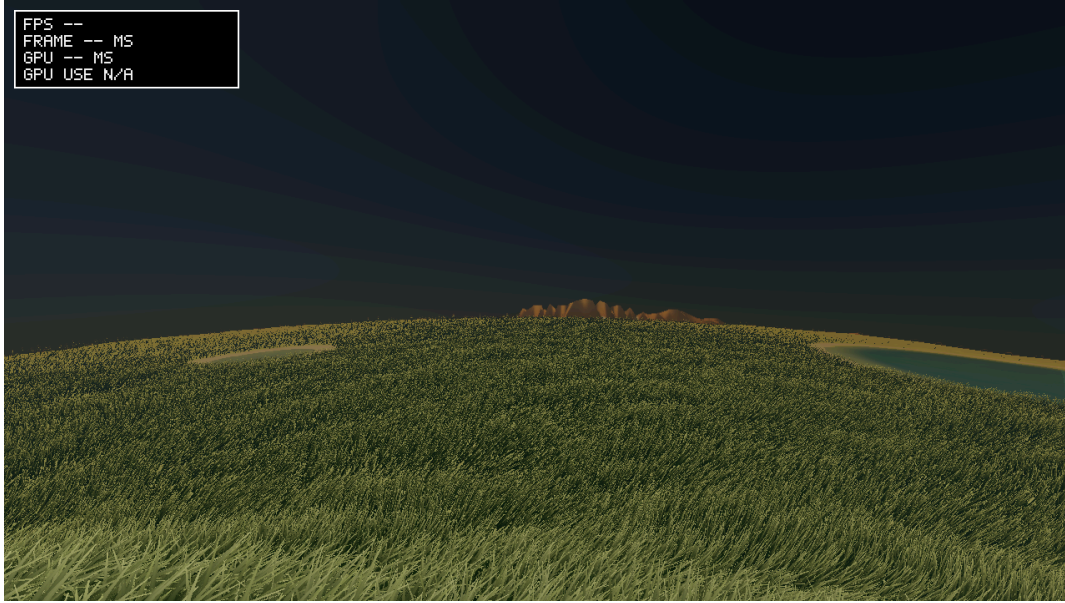


Figure 30: Actual current performance overlay under Xvfb/Mesa llvmpipe. The white-bordered black panel reports FPS, frame time, GPU pass time, and utilization availability; its numbers are capture-environment observations.

A best-effort 20 FPS quality target uses a smoothed frame estimate $m_t = 0.85m_{t-1} + 0.15f_t$ to select one of six scene-resolution scales, from 100% down to 25%. After a 1.5-second cooldown, sustained times above 55 ms lower resolution; times below 32 ms permit recovery. Only moving interactive scenes feed the controller. A graphics-memory estimate can cap the initial scale, reserving most reported memory for other uses; fixed-resolution captures remain deterministic. The HUD stays at native window resolution. Figure 31 shows the control law, not a measured response trace. This is a rendering control, not a change to the atmosphere’s physical parameters. Its lower resolution may still lose thin clouds or distant silhouettes; a strict 20 FPS guarantee is impossible once the minimum scale is reached or scene cost is unrelated to pixel count.

Best-effort quality control: timing feedback and memory cap

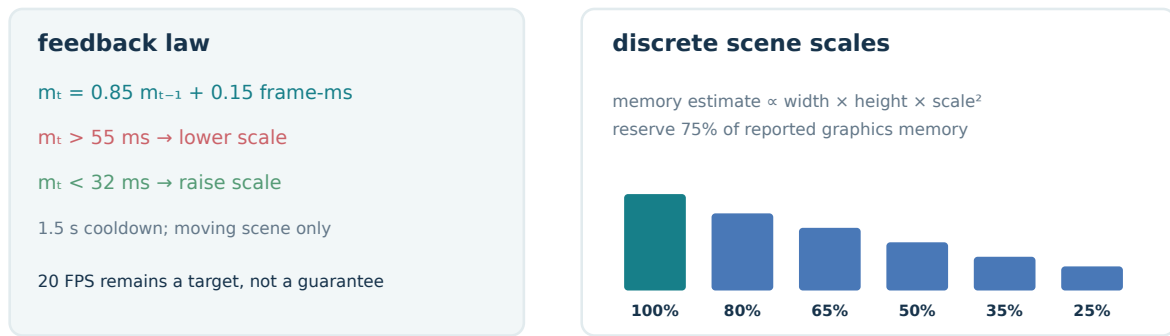


Figure 31: Actual controller thresholds and discrete scene scales; bar heights are relative scale, not measured frame rates. Generated schematic.

6.2. Visibility must follow the light path

The same visibility principle applies to air and water. A mountain blocks direct sunlight from nearby air, so the direct in-scattering term uses the planet's terrain depth map. The finite solar disk's twilight region remains, as does the approximate indirect fill. For water, a valid bright sample in the reflection image does not prove that a water point is lit: that point may lie on the night side or behind terrain. The direct reflection term therefore requires positive local Sun incidence and terrain visibility. Indirect water radiance remains. Figure 32 illustrates the distinction. The gallery's terrain-shadow and twilight cases are controlled visual regressions.

Why a geometric reflection must be gated on the night side

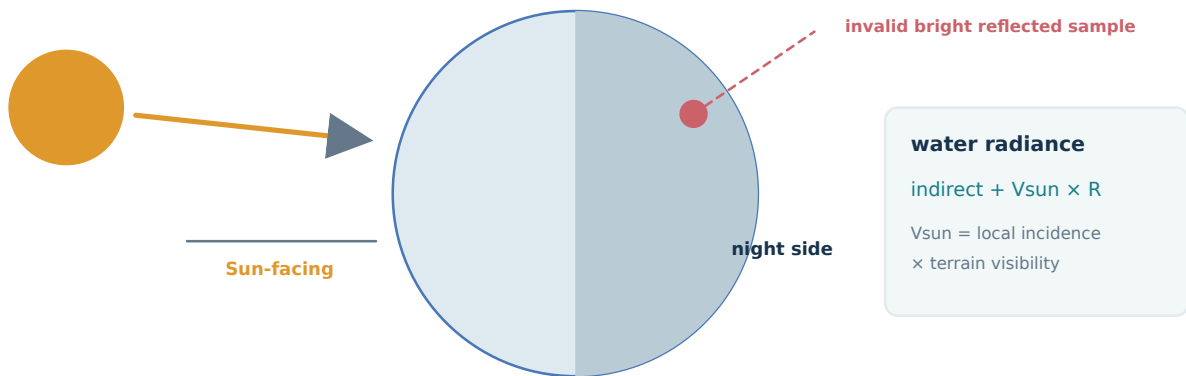


Figure 32: A reflected bright sample can exist in image space even when the water point lies on the night side. Local incidence and shadow-map visibility gate the direct reflection term. Generated schematic.

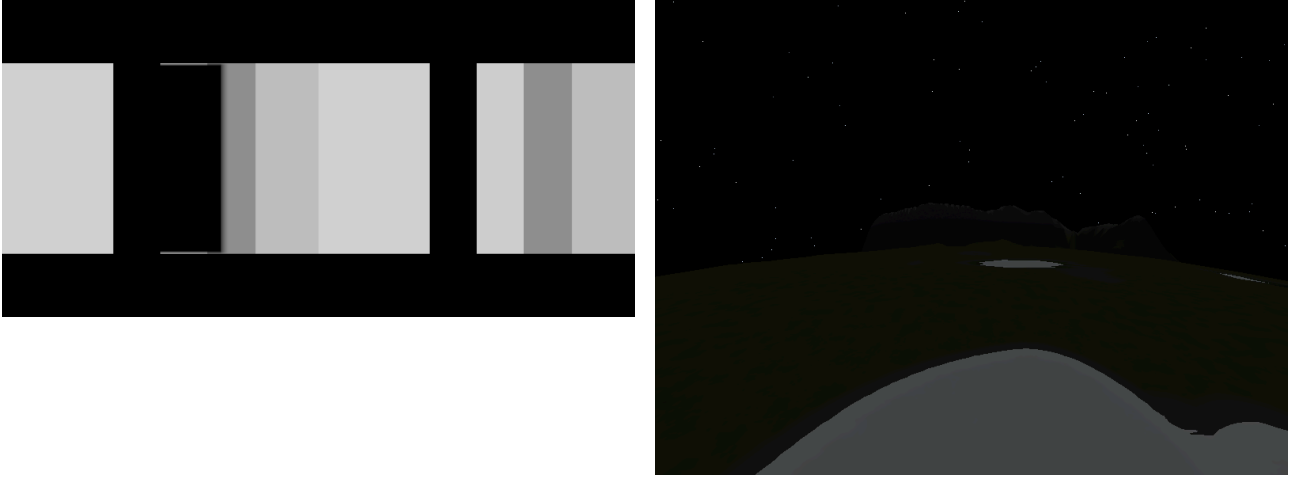


Figure 33: Actual current regression captures for terrain shadow visibility (left) and near-horizon twilight (right). Their fixtures and generation commands are recorded with the gallery.

6.3. Interpreting motion from the camera

Ten-revolution predictive trails and body cards help distinguish local orbits from inertial paths. The path color averages terrain vertex RGB factors multiplied by the body's configured base tint; labels report name, parent, axes, eccentricity, period, radius, and mass. Sampling the entire orbital system, rather than each child's local ellipse alone, is necessary for moving-parent trails. The overlay is toggled by 0 and hidden in planet and surface cameras; real X11 input tests verify both behaviors.

Orbit-to-surface entry is a one-second wall-clock transition. The starting pose is the last actually rendered view, avoiding a first-frame jump after the key event. An eased parameter $q(u) = 3u^2 - 2u^3$ drives spherical interpolation of the planet-relative radial direction, logarithmic interpolation of radius, quaternion interpolation of orientation, and field-of-view interpolation. Intermediate positions are clamped above sampled terrain and water. The target is resampled while the planet moves, so the transition ends at the current surface pose. This is a camera path rather than physically accelerated descent. The astronaut flight mode separately integrates forces, velocity and collisions. Figure 34 shows the path and ease curve.

One wall-clock second from orbital pose to surface pose

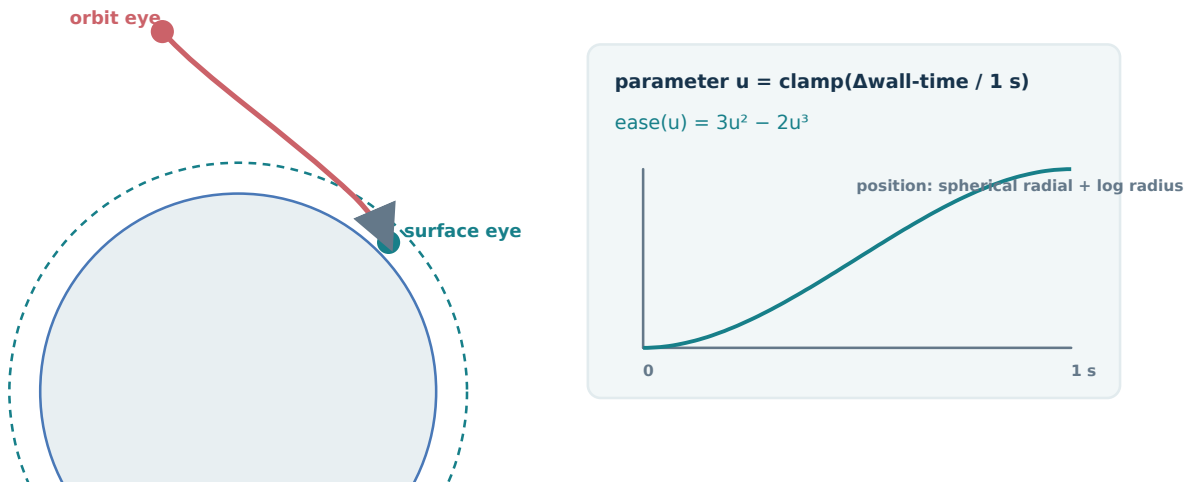


Figure 34: Orbit-to-surface descent. The curve shows the implemented smoothstep parameter; the spatial path is schematic because the actual start, target, and terrain vary by scene.

7. Astronaut locomotion and flight

Camera 4 follows an original procedural astronaut at the same surface location as camera 2. Normal movement is 6 m/s; either Shift key raises it to 12 m/s. The grey and blue suit carries small German upper-arm patches. Body-local grounded contacts follow planet translation and spin, while the grounded chase camera clears the ground and shortens its arm at intervening terrain. Airborne position and velocity use an inertial world frame.

An alternating gait preserves the stance foot and moves only the swinging foot, using a smooth ground-projected trajectory with 15 cm lift. Contacts intersect the installed terrain triangles, including LOD sinking, and resample

height after mesh replacement without changing radial location. The MIT-licensed ozz-animation 0.16.0 runtime supplies two-bone leg IK near the actor origin. This combines procedural step selection with a tested joint solver; it is an initial locomotion approximation rather than a motion-captured running cycle. Once walking settles, the pelvis rises over the locked ankles until both knees extend on level ground. The suit and chase view share this adjustment; uneven ground retains the necessary bend. The 0.10 s settling response preserves segment lengths, and exactly straight chains use a collinear midpoint to avoid an IK singularity. Replay includes the body offset; details and the before/after view are in `character/standing.md`.

Space launches at 10 m/s; holding it supplies upward thrust while airborne. A 100 kg actor integrates world-space gravity and quadratic drag relative to moving air, using configured pressure, gas composition and temperature. The suit tilts to direct its single downward exhaust axis, and arms remain quiet while the jetpack is armed. Falling has no artificial velocity cap; drag determines the local terminal speed. Landing resets the flight state. Flight uses full wall-clock elapsed time with 10 ms substeps and interpolated body motion, while grounded walking retains its presentation-time bound. The coasting exhaust axis remains inertial during planet spin.

The production main planet sizes a shared 8.29 kN engine, retained on other planets. At sea level, 100 m/s produces approximately 4.24 kN drag and 424 kW useful horizontal power. Assuming 1,000 m/s exhaust and 60% conversion efficiency, installed maximum thrust corresponds to about 6.91 MW exhaust input. These are explicit idealised choices, not measurements of a real jetpack. Attainable speed depends on gravity, air density and direction on the rapidly spinning miniature planet; the 100 m/s command changes force rather than clipping velocity. The reproducible calculation and limits are in `character/flight.md`. Fuel and thermal limits remain future work.

Airborne WASD now ignites thrust without Space and follows the full camera direction, including look-down descent; Space commands up. Within centre distance 1.2 diameters (2.4 radii), up follows the nearby planet or moon with a 0.35 s response. Beyond all such regions it remains inertial, with unrestricted space-camera pitch. The astronaut can approach the Moon and adopt its frame without teleporting or resetting velocity. Overlapping regions use 10% centre-distance hysteresis. Each flight step sums gravity from the three nearest celestial centres, including the Sun. Atmospheric drag is independent of the orientation switch. Space thrust requests acceleration without the near-body 100 m/s governor. Full details and world-state replay are in `character/space-flight.md`.



Figure 35: Actual renderer captures: planted feet and chase view, suit details and arm patch, tilted jetpack thrust and bubbles. Full pose and local effect phase are preserved in the replay sidecars.

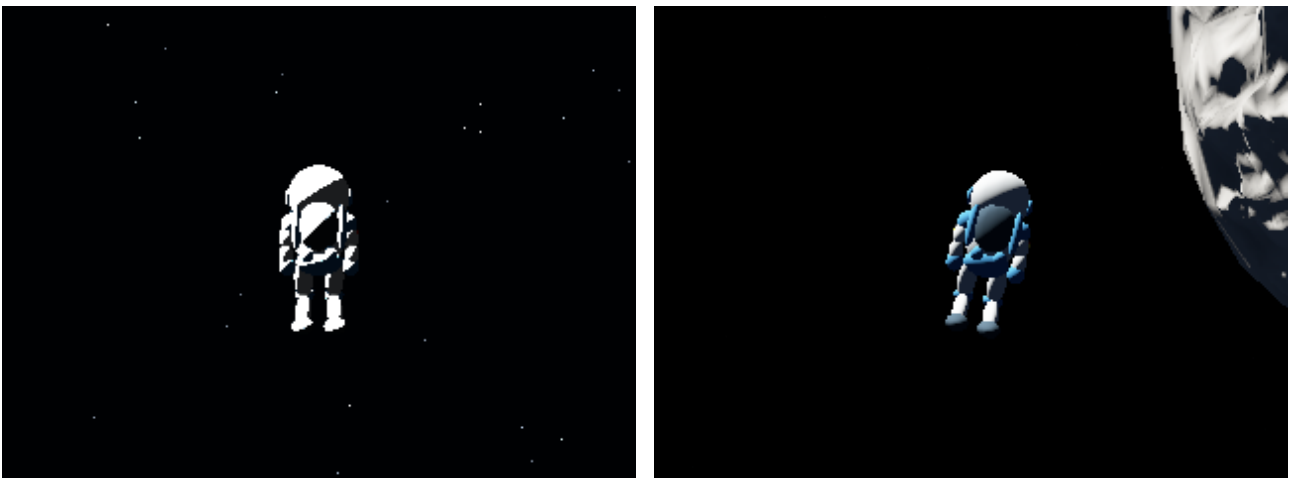


Figure 36: Reproducible seeded flight checkpoints: inertial space orientation (left) and entry into the Moon frame (right). These reduced-budget fixtures isolate the actor and omit foliage. Both PNG and complete pose replay exactly; the later settling check measures alignment to Moon up.

Exhaust markers now emit gradually at 40/s from alternating backpack nozzles, with a 64-particle cap and 1.4 s lifetimes. They inherit astronaut motion, relax toward moving air, expand and fade after thrust release; world

coordinates prevent relocation during a body-frame change. Grass and character effects share the same seeded Perlin gust, direction, flutter and local clock. Wind enters pressure-dependent astronaut drag and particle drift; grounded traction preserves planted feet. Rendering uses sorted instanced billboards, translucent spherical shells, Fresnel rims and Sun highlights, at most 128 triangles and 2,048 instance bytes per view. Main and water-reflection passes share the saved pool without advancing it. These are entrained visual markers, not a thermodynamic hot-exhaust solver. Details and replay history are in `character/exhaust.md`.

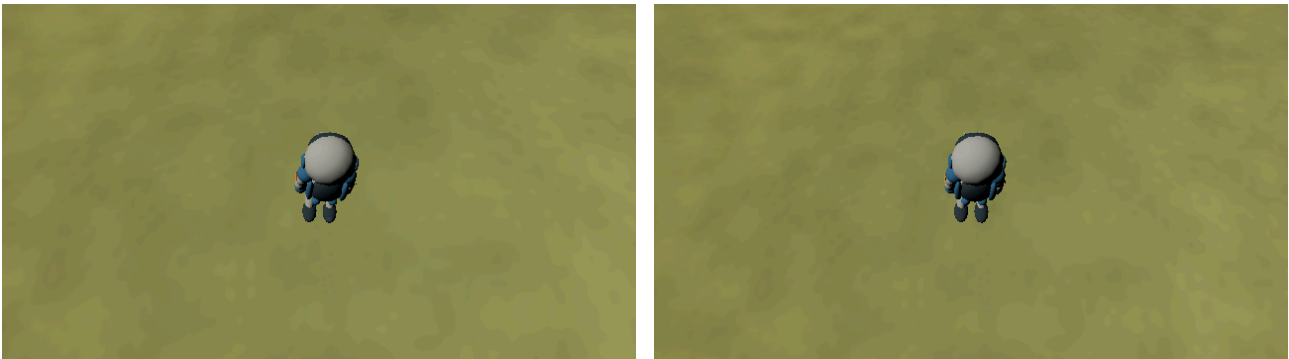


Figure 37: Persistent exhaust viewed from above: powered stream (left), released jetpack with its existing tail (right). The camera exposes the plume beneath the backpack; the reduced-budget fixture omits foliage while retaining its wind field. Both images replay byte-identically.

The actor receives terrain shadows and participates in HDR, atmosphere and water reflections. Local contact darkening helps anchor the feet; a full dynamic body shadow remains future work. CPU checks cover fixed stance contacts, joint reach, poles, mesh revision, chase clearance, 6/12 m/s movement, gravity and landing. GL captures check visible actor pixels and byte-identical gait and airborne replays; native input exercises camera switching, walking, jumping, boost and reload. These checks establish reproducibility and specific movement invariants on the tested renderer, not the naturalness of a sprint animation.

The requested Ava Turing model is not downloadable on Sketchfab. The catalog search favors Muko_Art's rigged stylized astronaut (11,718 triangles, one uploaded clip, CC BY 4.0), but its download needs authentication. Three public creator downloads are now prepared separately in `USER_I0/astronaut_vis`: AstroDev's astronaut (1,442 prepared triangles, 25 bones), Polygonal Mind's Astronaut 48 (3,478 triangles, 58 bones) and Square Cosmonaut 111 (4,506 triangles, 78 bones). The delivered versions have no backpack; two required removing the original pack. Sources, credits and separate CC0/CC BY licenses are retained. The accessible stylized models have less surface detail than Ava Turing. The game still uses its procedural character.

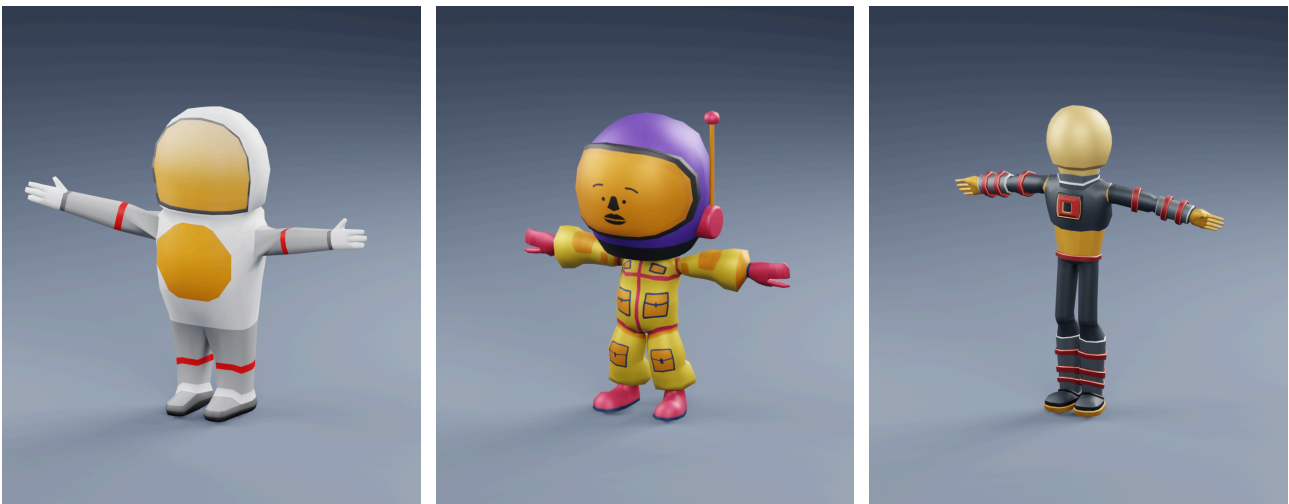


Figure 38: Downloaded candidates rendered locally from their prepared GLBs: AstroDev (left), Astronaut 48 (centre), Square Cosmonaut 111 (right). Original colors are retained; sources and modifications are credited in the candidate directory.

Each candidate retains connected thigh–shin–foot chains and normalized skin weights. An authored one-second FK inspection loop is sampled at 31 poses per prepared model; importing the exported GLBs checks another 17 poses each, finite bounded deformation and loop closure. Front, back and walking-pose PNGs show the delivered meshes. This demonstrates deformability, not a finished gait. Terrain stance locking, retargeting, skinning/material import and integration with ozz IK remain necessary for runtime use. Reproduction and evidence are in `character/assets.md` and `USER_I0/astronaut_vis/validation/`.

Grounded walking now leaves persistent grass trails. Body-local metre segments survive camera changes and planet transforms. A stackless radius-expanded BVH in an RGBA32F buffer texture selects nearby segments on both placement paths. Blades within 25 cm of the path centre lie at 8% height above the sloped walking path, extending 95% of their length along the walking direction; the effect fades smoothly to zero at 60 cm. Wind cannot lift centre blades again, while candidate roots, density, visibility and LOD remain unchanged. Airborne movement and water break path continuity, and relocation never draws a bridge between distant placements. Reload clears the history; a FIFO bound of 2,048 segments limits memory and eventually removes the oldest marks. This is geometric deformation, not a grass elasticity or recovery simulation.

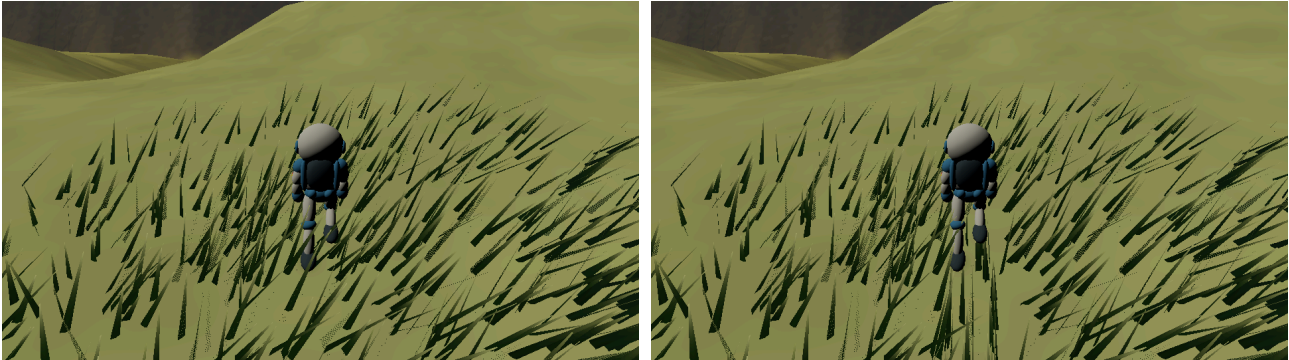


Figure 39: Identical pose, terrain, foliage roots and wind phase: unpressed control (left), persistent walking trail (right). The six-metre walk records 20 segments with 977 GPU-submitted blades; its replay is byte-identical.

Replay sidecars save the complete trail. CPU checks cover history bounds, discontinuities, invalid replay values and BVH distances at planet scale. Production-shader feedback checks fixed roots, flattened height, wind suppression, untouched surroundings, pole traversal, both geometry levels and both placement paths. Actual astronaut captures retain foliage, HDR and reflections and must replay byte-identically. Details and retained evidence are in `character/trails.md`.

The trail checkpoint combines a 52-entry full run (51 passed in 516.46 s, with RepositoryLayout waiting for the new gallery image) and targeted rechecks. Both final renderer/capture entries pass after the slope correction (122.35 s); RepositoryLayout passes after publishing the gallery. All 52 entries have passing results; this is a full run plus targeted checks, not a second clean full run. All 22 README images are regenerated and hash/fingerprint-verified, with the previous 21 PNGs unchanged. Evidence is retained under `character/trails/validation/`.

The offline checkpoint passes all 51 CTest entries in a clean 308.44 s run on GCC 12.2 / Mesa llvmpipe under Xvfb. Raw logs and compiler/executable identity are retained in `benchmarks/offline/validation/`.

8. Evaluation and model boundaries

8.1. What the evidence establishes

The matched physical-GPU comparison accepts twelve runs with unchanged source/config/binary hashes, verified EGL/NVML identities and 79 complete measured GPU samples per run. Camera and planned geometry/foliage receipts match all six pairs; culling differences and image errors are explicitly retained. At that comparison checkpoint, only benchmark tooling and documentation change from the preceding 63/63 runtime regression checkpoint. Repository checks and journal PDF review accompany the new raw evidence in `benchmarks/gpu-comparison/validation/`. These results establish workload-specific offscreen throughput, not CPU/compute migration acceptance or interactive FPS.

T3c5b1 passes all 64 CTest groups in one uninterrupted frozen-input run (1422.72 s), including seven controlled query cases and four actual-renderer frame cases. Quadro OpenGL 4.3 resolves 18 request-keyed work receipts and 963 frame spans with no dropped work, positive-timeout/flush fence polls, server waits or bulk buffer reads. Twelve EGL captures compare the prior binary, the new untraced binary and traced execution on Quadro and RTX, using CPU and compute terrain; each GPU/backend retains exact PNGs. The final source/build-input fingerprints and all 41 application/test executable hashes match frozen inputs after the full run; ten baseline and 22 gallery hashes remain unchanged. Evidence is in `architecture/terrain-gpu/async/hardware/timing/validation/`. These are bounded diagnostic/correctness checks on small hardware fixtures: native-loop wall time, publication latency/outcomes and physical-memory sampling remain T3c5b2, followed by matched hardware acceptance T3c5c. CPU terrain stays default.

T3c5b2a passes four relevant CTest groups in one final-input run (89.70 s): nine controlled timing cases, traced capture/replay, the new CPU/compute native loop test, and five existing native input scenarios/four startup rejection contracts. Both Mesa software and actual Quadro OpenGL 4.3 join 64 loop/frame observations through loading, rendered, minimized and resumed paths, without blocking GL fence polls, server waits, bulk buffer reads or explicit finishes. Private test delays establish scope correctness, not production frame cost. The old 60 s capture deadline was reached in preflight; a standalone recheck and the final 120 s allowance pass unchanged assertions. This is scoped validation, not a new full 65-group run. Frozen source/build-input and all 41 executable fingerprints, raw traces and preflight results are in `architecture/terrain-gpu/async/hardware/loop/validation/`. CPU default and T3c5b2b/T3c5b2c/T3c5c gates remain.

T3c5b2b1 passes ten relevant CTest groups in one final-input run (456.95 s), including five new controlled recorder cases, two real-worker ownership cases and two CPU/compute consumer-draw cases. Mesa and Quadro real GLFW fixtures join every successful publication to independently observed generations and frames, retaining zero blocking polls, server waits, bulk reads or explicit finishes. The initial private CPU draw helper assumed GPU consumer metadata; it now checks the managed flag, and the passing final run retains CPU/default compatibility. Source/build-input and all 41 executable fingerprints remain frozen; 22 gallery and ten retained historical baseline hashes match. This is scoped validation, not a new full 65-group claim. Evidence is in `architecture/terrain-gpu/async/hardware/publication/validation/`. Reload/capture/handoff/CPU grass-only tracing remains T3c5b2b2, physical memory T3c5b2c and hardware acceptance T3c5c.

T3c5b2b2a passes seven scoped CTest groups in 195.27 s, including 17 controlled timing cases and all four augmented resident reload cases. Evidence review found that the private reload fixtures did not bind GPU timing outside capture frames; adding explicit work bindings enables request-to-child dispatch/placement joins. Only the reload test executable changes, and focused software/Quadro rechecks validate the corrected fixture. The runtime source and the other 40 executable hashes retain their frozen fingerprints. Both renderers cover successful, invalid, superseded and preparation/final-exchange-failed reloads, matching final consumer generations without blocking GL polls, server waits or bulk reads; saved replay remains exact. This is scoped lifecycle validation, with a fixture correction and targeted rechecks, not a new full 65-group run or a hardware speedup claim. Twenty-two gallery and ten retained historical baseline hashes match. Evidence is in `architecture/terrain-gpu/async/hardware/publication/transactions/validation/`. Synchronous reload/capture/handoff/CPU grass tracing remains T3c5b2b2b, followed by physical memory and matched hardware acceptance.

T3c5b2b2b passes ten relevant CTest groups on final frozen inputs (366.53 s), including 23 controlled timing cases and capture/reload, CPU grass, Moon replay and native input coverage. Six actual Quadro OpenGL 4.3 checks pass. Both renderers retain nine capture reload roots/twelve children, join 57 GPU events including 30 off-frame events, verify CPU grass success/failure and one actual Moon handoff, and recover from seven injected reload fence failures. Replay capture replacing an exchanged child yields an incomplete root while replacement terrain publishes; fresh replay stays exact. Preflight exposed missing legacy grass request/generation keys and pending replaced children; final assertions validate the corrected identities and terminal outcomes. All 41 executables/source/build-test fingerprints remain frozen, and 22 gallery/ten historical PNG hashes match. Journal/evidence are in `architecture/terrain-gpu/async/hardware/publication/lifecycle/`. This is scoped correctness validation; physical memory T3c5b2c and matched hardware cost acceptance T3c5c remain, with CPU terrain the default.

T3c5b2c has passing results for twelve relevant CTest groups, including eight controlled reader cases and three memory renderer cases. The final runtime batch passes eleven groups in 429.91 s but exposes a fixture treating a valid empty native scene as invalid. A negative-radius input fixes that contract; the full renderer group passes in 67.55 s, with runtime source and the other 41 executables unchanged. Earlier validation exposed snapshot-lock loss of a lifecycle event, now covered by a deterministic held-lock regression and a separate bounded event queue; skipped latest state remains explicit. Three Quadro renderer cases and two native fixtures pass, with no frame memory queries or blocking GL work. Four small CPU/resident capture pairs on Quadro and RTX retain exact traced/untraced PNGs and match context/NVML UUIDs. Raw receipts, root joins and frozen 42-executable/driver-fixture hashes are in `architecture/terrain-gpu/async/hardware/memory/validation/`; 22 gallery and ten historical PNG hashes remain unchanged. This is scoped diagnostic correctness, not a new full-suite or production hardware cost claim. Device-wide samples may miss transient peaks, partial CPU vectors and process RSS overlap, and T3c5c remains with CPU terrain the default.

The exhaust/wind checkpoint passes all 56 CTest entries in a clean 675.37 s run on GCC 12.2 and Mesa llvmpipe/Xvfb with two rendering workers. Its ten new CPU cases validate emission, fading, bounded history,

inertial motion and wind drag; GPU checks validate transparency, Sun gloss, depth/state preservation and 120 production-GLSL Perlin comparisons. Burn, release and retirement retain exact images and complete effect/pose replay, alongside existing gait, flight, trails and Moon checkpoints. Native wind/character input, switching and reload pass. Evidence is in `character/exhaust/validation/`.

The space-flight checkpoint passed all 53 CTest entries in 764.20 s on GCC 12.2 and Mesa llvmpipe/Xvfb with two rendering workers. The character target includes 41 CPU cases, and native input verifies WASD-only airborne ignition. Flight/Moon images and their complete poses replay exactly. Evidence is retained in `character/space-flight/validation/`.

The repository contains 66 CTest entries covering configuration and orbital invariants; terrain, bounded worker ownership, camera and astronaut behavior; lighting, shadows, atmosphere and refraction on OpenGL; renderer captures; bounded timing diagnostics; adaptive quality; and real GLFW input under Xvfb. The 22 README images are checked against their SHA-256 manifest. These checks establish specific numerical and rendered behaviors on the tested software renderer. They do not establish physical fidelity, portability to every GPU driver, or a guaranteed 20 FPS for arbitrary scenes.

The directed-flight checkpoint passed 51 of 52 entries in 814.46 s. The nine-scene atmosphere test reached its 180 s deadline after eight scenes; with a bounded 300 s allowance, its standalone rerun passed in 152.20 s on the same application binary. All 52 entries therefore have passing results through full run plus targeted recheck, not a second clean full run. The motion target contains 19 passing CPU cases, and upright and tilted flight captures replay byte-identically. Raw evidence is retained in `character/flight/validation/`.

The standing update passes eight focused character/camera/renderer checks in 133.62 s, including 22 CPU motion cases and exact standing, gait, flight and trail replay. The two standing gallery views are newly posed; the other 20 captures retain their earlier provenance, with per-image source fingerprints for the updates. This is scoped validation rather than a fresh full 52-entry run. Comparison, legacy replay and artifact records are in `character/standing/`.

The October 2 continuation checkpoint passes all 49 entries in a clean 286.14 s run on GCC 12.2 and Mesa llvmpipe/Xvfb. The frozen grass view uploads 34,664 bytes of triangle IDs; cached frames upload zero grass-buffer bytes. Culling enabled/disabled and exact replay produce identical images. The October 1 paired compute/vertex benchmark remains a historical measurement on its original EGL host; this validation does not add a hardware timing claim. Evidence is retained in `benchmarks/procedural/validation/`.

The model boundaries are deliberate. Orbits are prescribed two-body ellipses with hierarchical recoil. Terrain is a deterministic synthetic height field with bounded tessellation. Atmospheric scale heights are chosen relative to shell size; the model does not solve hydrostatic temperature profiles, cloud dynamics, weather, or multiple scattering. Refraction bends view rays and reprojects available distant imagery but keeps near opaque geometry and direct shadow rays straight. Atmospheric temperature is read from JSON rather than evolved from sunlight and orbit. These limits define what a visual comparison can support.

T3b2 validates all 58 CTest groups across two final-executable runs: groups 1–29 before interruption and groups 30–58 in the resumed run. Four new allocation cases bring native terrain/metadata/allocation coverage to 14 cases. Compute and legacy replay, camera-only startup compatibility, scalar-only allocation readback, zero CPU render vectors, GL 3.3 fallback and native input pass. All 22 gallery hashes are retained. Evidence and the next asynchronous implementation plan are in `architecture/terrain-gpu/grass-allocation/` and `async/plan.md`. These software-renderer checks establish correctness, not hardware frame performance.

T3c1 passes eight controlled CPU cases and all 59 CTest groups in one uninterrupted final-binary run (736.20 s). The cases cover coalescing/backpressure, repeated epochs, body/field/mode/serial rejection, exception recovery, shutdown and independent resident snapshots. Compute captures pass worker-bound and transfer assertions; all ten compute/CPU/replay PNG hashes match T3b2 exactly. Lifecycle, reload and native input pass, and all 22 gallery hashes remain unchanged. Evidence is in `architecture/terrain-gpu/async/worker/validation/`. Interactive compute remains gated.

T3c2 passes all 59 CTest groups in one uninterrupted application-binary run (733.29 s), including 20 native compute cases. A test-only delayed-fence probe was then added; all 21 native cases pass on the rebuilt native target (24.294 s), while the application binary stays unchanged. Seven preparation cases cover chained submission, controlled zero-timeout polling, preallocated resources, admission/stale failure preservation, disabled consumers, malformed summaries and OpenGL state restoration. All ten compute/CPU/replay PNG hashes match T3c1 exactly and all 22 gallery hashes are retained. The capture fixture admits 4,632,148 logical bytes

and preallocates 524,320 draw bytes; its summary remains 224 bytes with zero metadata readback. Evidence for both native scopes is in `architecture/terrain-gpu/async/gpu/validation/`.

T3c3a passes 29 native compute cases (51.154 s), including eight complete-transaction cases. Six fence failures cover land, water, metadata, allocation, draw resources and retirement; all preserve live consumers and release staged buffers. Ten stale request variants, delayed zero-timeout preparation/retirement, actual old-resource draws, aggregate admission, externally changed live consumers and repeated body replacements pass. All ten compute/CPU/replay PNG hashes remain exact. Full regression, source/binary provenance and capture results are retained in `architecture/terrain-gpu/async/publication/validation/`. These are transaction API checks: renderer frame-boundary integration, shadow/cache refresh and transactional scene reload remain T3c3b/T3c3c.

T3c3b passes 31 native compute cases (52.055 s), including two new grass-only ownership/failure cases, and two renderer publication cases (13.408 s). Renderer checks observe matching main/shadow/reflection/water/grass identities after grass replanning, retain old buffers and the prior capture sidecar after an injected publication-fence failure, then recover in the same renderer. All ten compute/CPU/replay PNG hashes match T3c3a exactly; six character previews match their actual chase eyes, committed contacts and replay grass anchors. Full regression and frozen source/input/application/native/runtime provenance are retained in `architecture/terrain-gpu/async/renderer/validation/`. Scene reload transactions and asynchronous interactive opt-in remain pending.

T3c3c1 passes seven native scene-owner cases (25.031 s) and all 31 existing compute cases (48.264 s). Delayed bodies keep old terrain/water/grass drawable; preparation, scene-fence and retirement-poll failures retain prior resources and recover. Eleven changed request identities and superseded epochs never exchange a scene. Invalid configs, exclusive overlap and pre-dispatch admission fail without replacement buffers. Six replacements change body count/order and fields while preserving matching contacts and mounted cameras; old resources are released only after last-use retirement. Full regression and frozen source/input/application/native/runtime provenance are retained in `architecture/terrain-gpu/async/scene/validation/`. These are owner API checks; renderer reload and movement/Moon acceptance remain T3c3c2/T3c3c3.

T3c3c2 passes five actual renderer reload cases (58.680 s), nine CPU worker cases (0.085 s), seven owner cases (33.306 s) and all 31 native compute cases (73.731 s). Changed body counts, fields and cameras match fresh captures byte for byte. All six per-body preparation fences and the final scene fence preserve old scene/buffers on failure; generated generation-owned staging buffers are released and the same renderer recovers. The probe reuses the bounded field-parameter cache. CPU/legacy invalid-config recovery, optional-camera removal and repeated orbit reloads pass. Third-person replay keeps exact terrain/grass anchors and images despite a file change after staging; malformed trail/exhaust, changed pipeline and unsupported versions retain the current scene. All 59 CTest groups pass in one uninterrupted final-binary run (1051.70 s); ten compute/CPU/replay PNG hashes match T3c3c1 exactly and 22 gallery hashes remain unchanged. Frozen source/input/binary and artifact evidence are in `architecture/terrain-gpu/async/reload/validation/`. Movement/body-switch/Moon acceptance remains T3c3c3 and interactive compute remains gated.

T3c3c3 passes three actual renderer recovery cases (73.010 s), eleven CPU worker cases (0.084 s) and existing CPU space-flight/Moon exact replay (42.37 s). Thirteen-frame continuous 12 m/s captures cross terrain rebuild thresholds, retain instantaneous pose/old consumers after invalid config and GPU failure, then recover through three body reorders and field changes. Ready old results are superseded during failed and successful replacements; controlled running/queued and ready/queued cases prove bounded stale-work disposal and continued live-epoch use on the same executor. Moon arrival binds destination contacts at revision 2 on the handoff frame, replacing the old revision-0 gap. Saved Moon reload, repeated reload and fresh replay retain exact PNGs. Main/shadow/applicable reflection/water/grass/contact keys and revisions match; old buffers stay alive until retirement and are deleted before later allocations reuse names. Observed logical overlap peaks at 13,179,460 bytes under the existing limits; it excludes total CPU/driver VRAM. All 60 CTest groups pass across an interrupted run and resumption on frozen inputs (952.49 s summed passing-group time); ten baseline and 22 gallery PNG hashes remain unchanged. Frozen source/input/five-binary and artifact evidence are in `architecture/terrain-gpu/async/recovery/validation/`. T3c3 is tested; T3c4 asynchronous interactive opt-in and T3c5 hardware acceptance remain pending. CPU stays default and interactive compute gated.

T3c4a passes three actual GL frame cases (20.148 s), eleven worker cases and all 61 CTest groups in one uninterrupted frozen-input run (927.29 s). Startup, programmatic 6/12 m/s movement, grass-only failure/recovery, delayed/failed retirement, stale masks and retry backoff retain matching consumer keys. No positive-timeout/flush fence polls, server waits or bulk buffer readbacks are observed. Logical overlap reaches 8,408,696

bytes, excluding physical VRAM and CPU snapshots. Ten baseline and 22 gallery hashes remain unchanged; source/input/six-binary provenance is retained in `architecture/terrain-gpu/async/interactive/validation/`. Resume asynchronous reload T3c4b; native input/public CLI acceptance T3c4c and hardware gates T3c5 remain.

T3c4b passes four actual GL asynchronous reload cases (26.511 s), fourteen worker cases and all 62 CTest groups in one uninterrupted frozen-input run (884.40 s). Old-scene 6 m/s movement and managed draws remain usable during delayed preparation; executing/queued/ready future work is superseded without joins. Body reorder/count/field changes, invalid config/replay, GPU and final-exchange fence failure, delayed/failed retirement, startup requests and optional-camera/empty-scene recovery pass. Saved replay recovery is PNG-exact. Zero positive-timeout/flush fence polls, server waits and bulk readbacks are observed. Captured logical accounting peaks at 16,300,856 bytes; CPU snapshots and driver physical VRAM are excluded. Ten baseline and 22 gallery hashes remain unchanged; source/input/seven-binary provenance is retained in `architecture/terrain-gpu/async/interactive/reload/validation/`. Native compute input/public opt-in remains T3c4c, hardware cost T3c5 and CPU the default.

T3c4c passes five production native GLFW/X11 processes and four shipping startup rejection contracts (48.09 s focused), and all 63 CTest groups in one uninterrupted frozen-input run (924.49 s). Native standing, 6/12 m/s walking and sprinting, jump, directional/Space thrust, trails and exhaust retain matching draw/contact receipts. Saved initial outer-space/Moon starts use actual native input and replay reload, with inertial up/three gravity sources and same-frame Moon contact binding. R/file-watch reload, old-scene movement during delayed preparation, invalid config/GPU failure, supersession/body reorder and delayed/failed retirement recover. Loading input exposed and now covers the profiler scope crash. Zero blocking/flush fence polls, server waits, bulk readbacks and explicit finishes are observed. Native logical accounting peaks at 14,348,020 bytes, excluding CPU snapshots and driver physical memory. Ten baseline and 22 gallery hashes remain exact; frozen source/input/eight-binary provenance and complete compressed native traces are retained in `architecture/terrain-gpu/async/interactive/native/validation/`. Explicit interactive compute requires resident GPU grass; CPU default, fresh GL 3.3 fallback and locked replay rejection remain. T3c5 hardware total-frame and physical-memory acceptance is pending.

A1a and T3c5a pass all 63 CTest groups in one uninterrupted frozen-input software run (1005.40 s), including five new cache cases and an actual-GL uniform check. Reference packs are reused across moving/paused and main/reflected views; optical input changes and invalid refresh/reload retention are tested, and CPU/compute replacement captures match fresh images. Quadro atmosphere/capture checks pass in 14.007 s and optical reload parity in 8.945 s. Hardware native acceptance passes five input scenarios and four startup rejection contracts in 31.594 s. NVIDIA's exact GL 3.3 response exposed a false compute fallback; production windows now prefer GL 4.3 and retry GL 3.3. Complete traces identify Quadro GL 4.3 for native input/space/Moon/default CPU and Mesa GL 3.3 for fallback, with zero blocking polls/server waits/bulk buffer reads/finishes. Source/input/ten executable fingerprints, exact nine regenerated atmosphere PNGs, ten baseline PNGs and 22 gallery hashes are retained in `architecture/atmosphere/optics/validation/`. RTX 3070 Ti is visible through NVML but its OpenGL context is not tested. T3c5 matched hardware total-frame/publication/physical-memory measurements and A1b GPU highlight reduction remain pending; CPU stays default.

T3c5c1 passes two scoped native CTest groups in 63.63 s and three actual Quadro GLX production pairs. Independent archived rechecking joins all 1,440 measured and 1,955 excluded native frames, 3,494 ready GPU work rows and twelve published initial terrain attempts, with exact paired field/topology, pose and planning anchors. Every measured frame has a ready GPU span; no measured outlier is removed or blocking GL work introduced. Runtime/application and the other 41 executables remain unchanged; frozen source/input/42-executable hashes and full compressed traces are retained in `architecture/terrain-gpu/async/hardware/cost/preflight/validation/`. Gallery/historical captures stay unchanged. Movement, rendered coverage and replacement/final gates remain pending.

T3c5c2a has passing scoped results for seven relevant CTest groups. Six pass in the initial 217.51 s batch; an augmented native input group passes in 92.11 s after correcting the short-input sample window and separating a warmed flat/coarse clock fixture without foliage. The original noisy input/reload and space/Moon checks remain intact. Software CPU/compute/GL 3.3 fallback wall speeds are 5.965–6.003 m/s walking and 11.886–12.011 m/s sprint; three >1 s frames retain bounded continuous walked history. Runtime and all 42 executables stay frozen through the Python fixture correction. Failed preflights are retained. Twelve before/final/delayed speed-gate runs and four retained short preflights join 933 movement and 4,231 excluded frames. Three fresh stationary pairs retain 1,440 measured and 1,942 excluded frames, 3,486 ready GPU work rows and twelve

complete publication receipts with exact paired geometry/pose/anchors. Native p95 ratios are 1.0368, 1.0227 and 1.0853: the third pair exceeds the 5% migration limit, so complete cost/stage investigation and CPU default remain. Gallery and historical captures stay unchanged. Full compressed traces and before/after provenance are in `architecture/terrain-gpu/async/hardware/cost/movement/validation/`. This is scoped validation, not a new full-suite or migration cost claim.

T3c5c2b1 independently verifies twelve fresh production native 400 m routes and all distance-cohort joins from compressed archives. Actual speed, fixed quality/budgets, device UUID, bounded worker state and compute draw/contact keys pass; every measured GPU frame and 12,798 full-run GPU work rows are ready. All 511 publication outcomes, memory age/status rows and 192 landmark brackets remain. The first complete CPU preflight exposed an analyzer assuming a managed memory ledger on the unmanaged CPU path; corrected null semantics are checked against that retained real trace, and the fresh paired runs use the frozen corrected scripts. Source/runtime, all 42 executables and driver fixtures remain unchanged from T3c5c2a. No new runtime build, CTest suite or coverage acceptance is claimed. Gallery/historical image hashes are retained. Independent speed/p95/bin/landmark/quality/outcome/memory checks and artifact digests are in `architecture/terrain-gpu/async/hardware/cost/routes/validation/`.

T3c5c2b2a passes one new scoped inspection group twice (22.93 s and 28.56 s), including reused-output readiness. Eight CPU/compute airless/HDR native Quadro snapshots and two excluded production sprint snapshots also pass. The independent archive validator recomputes all 26 final raw queue/depth/frame joins, masks, distance-band counts and device identities. An 8.00 s qualification rerun exposed stale snapshot files after an earlier successful pass; atomic JSON publication, removal of stale request metadata and current-frame joins fix the fixture. Failed mixed old/new preflight evidence remains separate. Product source/shaders and all 42 prior executables/driver fixtures remain unchanged; only the new private inspection binary and test inputs are added. This is scoped probe qualification, not a new full suite or rendered-density/cost acceptance. Gallery/historical hashes remain unchanged. Evidence is in `architecture/terrain-gpu/async/hardware/cost/coverage/validation/`.

T3c5c2b2b passes two scoped software groups in 44.98 s and eight matched Quadro fixture snapshots. Independent archived checks verify 48 final native snapshots, including 40 controlled snapshots, raw plane/area/root/final-compositing joins and the three stationary repeat pairs with exact geometry/pose/anchors. Import/path/NumPy launcher failures and a superseded pixel-center preflight remain separate. The comparison study records four declared coverage failures and analytic estimator sensitivity; methodology qualification and fresh repeats remain. Product source/shaders, app/timed probe and 42 prior executables/driver fixtures stay unchanged; all 43 final executable/input hashes remain frozen through measurements. Lossless XZ storage of 52 large ground maps saves 187.4 MiB, with decoded lengths/hashes equal to original readbacks. Gallery/historical captures stay unchanged. This is scoped inspection/archive qualification, not a full-suite, true-area or migration-speedup claim. Evidence is in `architecture/terrain-gpu/async/hardware/cost/coverage/matched/validation/`.

T3c5c2b2c1 passes one scoped software CTest in 53.80 s and the same 16 GPU snapshots on verified native Quadro OpenGL 3.3. Independent raw reconstruction checks all 32 final snapshots against the explicitly reused exact polygon/circle oracle, readback/resource bounds and unchanged declared gates. Both failed qualification batches, their float maps/inputs and all final evidence are retained losslessly in `architecture/terrain-gpu/async/hardware/cost/coverage/area/validation/`. Product sources/shaders and all 43 previous executables remain unchanged; only the private area probe and test inputs are added. This is known-geometry measurement qualification, not a full CTest suite, physical-memory peak or live route acceptance.

T3c5c2b2c2 passes three frozen scoped software groups in 161.68 s and eight native Quadro fixture snapshots, plus four excluded production inspections. Raw reconstruction verifies 64 queue/depth snapshots, 28 matched scene receipts, 60 live sampling grids and 180 biome diagnostics, explicitly reusing the frozen qualified analytic accumulator. Complete final/preflight payloads and unchanged shipping/43-other-executable provenance remain in `architecture/terrain-gpu/async/hardware/cost/coverage/live/validation/`. This is live inspection qualification, not a full suite, physical-memory measurement, fresh three-pair coverage or migration acceptance.

T3c5c2b2c3 passes four driver checks and two repeated software groups (126.34 s), plus eight requalification snapshots on verified Quadro. Independent reconstruction checks 40 final live inspections /120 sampling grids /360 biome diagnostics and 80 native/controlled queue-depth snapshots, including all 24 fresh route crossings and twelve pair decisions. The strict projection assertion now derives clip provenance from the renderer's actual double-precision chase eye; qualified raster/area math and shipping source are unchanged, with only the private coverage probe rebuilt. Initial projection, slow storage and manifest-order packaging preflights remain lossless. Exact recovered preflight source bytes reproduce both frozen build/test tree hashes.

Runtime/43 other executable provenance and all final inputs remain frozen. This is scoped coverage verification, not a full suite or timing/migration acceptance.

T3c5c2b3 passes the scoped software raster-control CTest in 69.00 s and the same 27 controls/108 frames on verified Quadro. Legacy compute placement, resident planning and vertex placement use independently queried blade/placement uniforms, generated primitives and passing depth/stencil samples. Fixed/indexed full/discard primitive counts match; discard samples and suppressed primitives/samples are zero. These blocking checks are excluded from timings and do not measure fragment shader invocations or production density. Independent archive reconstruction verifies all 54 runs, 27 pairs, native publication/wait/readback audits and every stage distribution. All 703 source payloads match after lossless decoding (152,526,975 raw bytes to 7,740,877 stored). Successful label/count preflights and an archive string/Path adapter failure remain separate. Shipping/shaders and 44 previous executables are unchanged; 45 final executable/input hashes stay frozen. This is scoped control/archive verification, not a full suite or accepted hardware speedup. Full-draw failures remain.

The private blade-order software CTest for T3c5c2b4 passes in 35.27 s. All 18 final and 54 preflight snapshots are independently decoded and rechecked. Main detailed/quad counts are 772/396 and reflection counts 11/166 on both backends; exact multisets, matrices and depth yield zero displayed RGB differences across permutations. Double contact root drift up to 1.421×10^{-14} m is reported; exact float roots and native cameras are retained. An initial overly strict double-root archive assertion, coarse empty-quad fixture, first refined run and preflight before per-frame exclusion flags remain separate. All 1,675 original payloads match after lossless decoding (110,730,278 raw to 32,018,882 stored bytes). Shipping/shaders and 45 prior executables remain unchanged; 46 final input/executable hashes are frozen. These are software control prerequisites; the archived GPU device/context failures prevent production diagnosis or hardware cost acceptance. No full-suite run or runtime optimization is claimed.

8.2. Rebuilding the paper and the evidence

From the repository root, regenerate the deterministic vector figures and compile the Typst source:

```
python3 docs/journal/make_figures.py
typst compile --root . docs/journal/paper.typ docs/journal/paper.pdf
```

The 12 SVGs require only Python's standard library. The paper references the versioned PNGs already stored in docs/screenshots/. To rerun the application tests in a headless OpenGL environment, build the CMake project with testing enabled and execute `LIBGL_ALWAYS_SOFTWARE=1 xvfb-run -a ctest --test-dir build --output-on-failure`. Exact image-generation commands and checksums are in docs/captures/generation.json.

9. Conclusion

The renderer's most useful physical separation is between the state of a world and the approximation used to draw it. A fixed terrain height field can be sampled at changing mesh densities; prescribed orbital positions can be evaluated at any time; and atmospheric extinction, in-scattering, and refractive curvature can be varied from a configured air state. The atmosphere is strongest as an interactive optical explanation of why density, path length, particle type, and visibility change a sky. Its limits are equally specific: shell-relative scale heights, RGB coefficients, single scattering, fixed reference weather, and image-based refraction prevent a predictive climate or precision optical interpretation.

10. References and implementation map

R1. NASA, *Basics of Space Flight, Chapter 3: Gravity and Mechanics*, official orbit primer. Used for the classical two-body context; the implementation details above come from `src/simulation/OrbitalSystem.h`.

R2. E. Bruneton and F. Neyret, "Precomputed Atmospheric Scattering," *Computer Graphics Forum* 27(4), 2008, doi:10.1111/j.1467-8659.2008.01245.x. The project uses a smaller runtime single-scattering approximation and a limited reusable lookup.

R3. NIST Engineering Metrology Toolbox, Index of Refraction of Air documentation. Background for pressure, temperature, humidity and refractive index; this project does not claim NIST-grade refractometry.

R4. IAU 2015 resolutions, official PDF. The rendering code uses the Resolution B2 bolometric zero-point convention for luminosity.

R5. *PlanetSimulation* repository: implementation in `src/`, shaders in `shaders/`, checks in `tests/`, and captured images with reproducibility metadata in `docs/captures/generation.json`.

R6. NASA Glenn Research Center, Ideal Gases under Constant Volume, Constant Pressure, Constant Temperature, and Adiabatic Conditions. The hydrostatic, isothermal scale-height derivation is the physical motivation for the exponential profile, not the formula used to set the renderer’s shell-relative scale heights.

10.1. Claim-to-artifact map

Claim	Primary implementation	Validation
Matched physical-GPU throughput	<code>scripts/benchmarks/gpu_compare/</code>	Twelve runs, UUID checks, complete queries and paired images
Complete native loop wall	<code>diagnostics/timing/NativeLoopProfiler.h</code> , <code>Interactive.cpp</code>	Controlled clock cases and real GLFW loading/minimized/resume
Complete publication lifecycle	<code>diagnostics/timing/PublicationProfiler.h</code> , <code>worker/reload/draw endpoints</code>	Controlled scopes, terminal families, capture/grass anchors and actual destination contacts
Production stationary workload preflight	<code>scripts/benchmarks/terrain_cost/</code> , private native probe	Three CPU/compute pairs; exact geometry/pose/anchors, complete GPU and memory receipts
Native real-time grounded speed	<code>Interactive.cpp</code> , private native input/speed tools	Production monotonic speeds, delayed frames, suspension bound and fresh stationary controls
Production native route cost preflight	<code>scripts/benchmarks/terrain_cost/routes.py</code> , archive validator	Twelve 400 m runs, distance cohorts/landmarks, full raw outcomes; rendered coverage pending
Live native grass inspection	<code>tests/app/terrain/native/coverage/</code>	26 final root/depth/frame snapshots; excluded blocking reads; matched density pending
Matched live-plan coverage comparison	<code>coverage/matched/</code> , private prepare/draw wrappers	24 excluded routes, 48 native /40 controlled snapshots; estimator sensitivity and stationary failures retained
Analytic area measurement prerequisite	<code>coverage/area/</code> , tiled private raster probe	32 known-geometry GPU snapshots, exact band references and 1% convergence
Live analytic inspection qualification	<code>coverage/live/</code> , installed-VAO raster and excluded analysis	20 snapshots, 60 grids, true opaque occlusion, 180 biome checks; fresh route acceptance pending
Fresh analytic route comparisons	<code>coverage/repeats/</code> , fixed qualified area and native plans	24 routes /12 passing pairs; crossing brackets, root variance and display-mask uncertainty; timing pending
Stationary wind/raster investigation	<code>cost/raster/</code> , private shader/draw wrappers	54 runs /27 pairs; qualified controls, retained full-draw failures and raster-enabled localization
Excluded blade-order controls	<code>cost/raster/order/</code> , private queue permutations	18 software snapshots preserve contents/matrices/depth/color; physical-GPU diagnosis blocked by device access
Device-verified memory	<code>diagnostics/memory/</code>	UUID/error/bounded-reader cases, GPU and native receipts
Bounded asynchronous GPU timings	<code>diagnostics/timing/</code> , <code>FrameProfiler.h</code>	Controlled query tests and traced resident publication
Stable terrain under LOD	<code>src/rendering/geometry/Terrain.h</code>	<code>TerrainTests</code> , surface capture
Bounded CPU terrain jobs	<code>src/rendering/geometry/jobs/</code>	<code>TerrainJobTests</code> , compute capture
Staged GPU preparation	<code>src/rendering/geometry/compute/</code> , <code>foliage/procedural/</code>	<code>GpuPreparation</code> native cases, compute capture
Complete GPU transactions	<code>src/rendering/geometry/publication/</code>	<code>TerrainPublication</code> , renderer publication/reload recovery
Renderer terrain consumers	<code>src/rendering/runtime/terrain/</code> , <code>ScenePass.cpp</code>	<code>TerrainPublicationRenderer</code> , compute capture identities
Resident asynchronous frames	<code>src/rendering/runtime/terrain/</code> , <code>InteractiveTerrain.cpp</code>	<code>TerrainFrameIntegration</code> , zero-timeout polls and matching draws

Claim	Primary implementation	Validation
Asynchronous complete reload	src/rendering/runtime/terrain/reload/interactive/	TerrainReloadFrameIntegration, future-worker lease tests
Native compute opt-in	src/app/, src/rendering/runtime/Interactive.cpp	TerrainNativeInputIntegration, X11 input and zero-wait audit
Whole-scene exchange owner	src/rendering/runtime/terrain/reload/	SceneReplacement, RendererReload, TerrainRecovery
Barycentric orbit hierarchy	src/simulation/OrbitalSystem.h	OrbitalSystemTests
Shadowed atmosphere	shaders/atmosphere/atmosphere.frag	TerrainShadowRenderIntegration
Shared reference optics	src/rendering/atmosphere/AtmosphereOpticsCache.h	AtmosphereTests, GL bindings, captures and renderer reload
Temperature-controlled bend	src/simulation/Atmosphere.h	AtmosphereTests, GPU render test
Adaptive scene scale	src/rendering/diagnostics/AdaptiveQuality.h	AdaptiveQualityTests, live render
One-second descent	src/rendering/camera/CameraTransition.h	CameraTransitionTests, X11 input
Planted feet and local flight	src/rendering/character/	AstronautMotionTests, capture and X11 input